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To cite this article: Anouk Dieuleveut & Ira Noveck (07 Jul 2026): Learning modal force: Evidence from French, Language Acquisition, DOI: [10.1080/10489223.2026.2675468](https://doi.org/10.1080/10489223.2026.2675468)

To link to this article: <https://doi.org/10.1080/10489223.2026.2675468>



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Published online: 07 Jul 2026.



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Learning modal force: Evidence from French

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ABSTRACT

In English or French, modals express either possibility (e.g., “you can come”) or necessity (e.g., “you must come”). Prior acquisition studies have shown that English-speaking children master possibility modals early, but have difficulties with necessity modals, based on converging evidence from both comprehension experiments and corpus studies of children’s spontaneous productions. However, the underlying cause of this *Necessity Gap* remains debated. One issue is that prior studies have focused on English, a language in which necessity modals are much rarer than possibility modals in children’s input, which could participate in their later mastery. In this study, we test the hypothesis that children’s struggles with necessity modals are due to limited exposure by investigating how children learn modals in French, a language where necessity modals are more frequent in parental speech. We present a detailed corpus analysis of 2- to 6-year-old French children’s modal productions and their input as well as evidence from two experimental studies based on the corpus. We show that despite their higher exposure to necessity modals, French-speaking children experience the same difficulties as their English-speaking peers: They start producing necessity modals later, use them less frequently than possibility modals and, crucially, do not use them in an adult-like way. Additionally, we show that these difficulties persist up to the age of five. We conclude that the delay found in English learners cannot be explained by limited exposure and discuss various hypotheses to explain children’s struggles: conceptual, semantic, or pragmatic.

ARTICLE HISTORY

Received 13 December 2025

Accepted 11 May 2026

1 Introduction

Human language allows us to talk about events and situations that are not actual, and may or may not ever become so. We can describe what we are able to do given our physical capacities (e.g., “I can run a marathon”), what we are allowed or required to do given certain rules (e.g., “in this city, cyclists must wear a helmet”), what we should do in order to reach our goals (e.g., “to get good bread, you have to go to a French bakery”), or what is more or less likely to be the case given the available evidence (e.g., “it is late, you must be hungry”). Modals are words, such as *can* or *must*, that allows us to talk about possibilities and necessities.

The question of the emergence and early development of modal language – when do children start to use linguistic expressions that refer to non-actualities? How early do they understand what these expressions mean? – has been a major topic of inquiry since the eighties, with renewed interest in recent years (e.g., Cesana-Arlotti et al. 2018; Leahy and Carey 2020; Leahy and Zalnieriunas 2021; Grigoroglou and Ganea 2022; Phillips and Kratzer 2024). In this paper, we focus on a specific question within this debate: when and how do children figure out the “force” of modals, that is, that words such as *pouvoir* “can” express possibility (1a), whereas words such as *devoir* “must” express necessity (1b)?

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(1)	a. Tu	peux	venir.	POSSIBILITY
	You	can.2SG.PRS	come.INF	
	'You can come'			
	b. Tu	dois	venir.	NECESSITY
	You	must.2SG.PRS	come.INF	
	'You must come'			

Sentence (1a) means that it is possible to come, but there are other options available. Sentence (1b) has a logically stronger meaning: that it is necessary to come, with no other options available. From a logical standpoint, necessity (e.g., (1b)) entails possibility (e.g., (1a)). This entailment relationship creates a complex learning problem for the child: given that every time one hears sentences such as (1b) used, the corresponding possibility statement (1a) is also true, what prevents learners from assuming possibility meanings for necessity modals? This kind of *subset problem* arises in any case of word learning involving a set/subset relationship—such as *apple* versus *fruit*. With modals however, the learning task is further complicated by two aspects: first, their meanings might be less directly “observable” than content words; second, they allow to convey various types (or “flavors”) of modality.

Prior acquisition research has established that there is an asymmetry in English-speaking children’s acquisition of possibility versus necessity modals: children master possibility modals early, but struggle with necessity modals. This conclusion is based on converging evidence from both behavioral experimental studies and corpus analyses of children’s natural productions. Behavioral experiments, testing children’s comprehension, have shown that English-speaking 4- to 5-year-olds tend to both over-accept possibility modals in necessity situations, and to accept necessity modals in possibility situations (Noveck et al. 1996; Noveck 2001; Ozturk and Papafragou 2015; Moscati et al. 2017, Leahy and Zalnieriunas 2021; Staniszewski et al. 2023; Bleotu et al. 2025; Cournane et al. 2025). Corpus-based studies, focusing on children’s spontaneous productions, have shown that English-speaking children start producing possibility modals before necessity modals, use them more frequently, and use them in an adult-like way, whereas they do not use necessity modals in appropriate conversational contexts (Dieuleveut et al. 2019, 2022). Finally, elicited production studies have shown that English 3–4-year-olds have a general preference for possibility modals, and when using necessity modals, they use them indiscriminately across necessity and possibility situations (Cournane et al. 2024). However, the source of these difficulties with necessity modals—what has been called the *Necessity Gap* (Dieuleveut 2021)—is debated. Hypotheses put the burden on conceptual, semantic, or pragmatic levels.

One issue is that we do not know whether these difficulties are specific to English or cross-linguistically general, because all these studies have focused on this language. This is a serious concern as it has been shown that in English children’s input (i.e., the parental speech they are exposed to), possibility modals are considerably more frequent than necessity modals: *can*, the English most common possibility modal, occurs three times more often than *have to*, the most common necessity modal (Dieuleveut et al. 2022). This lower frequency of necessity modals in children’s input could explain the asymmetry and the “delay” with necessity modals, if frequency of exposure facilitates word learning as is often assumed (see Ambridge et al. 2015).¹ To assess whether the *Necessity Gap* represents an issue beyond frequency, we need to look at modal acquisition in languages without the same input asymmetry between possibility and necessity.

¹High-frequency words are acquired earlier than low-frequency ones (e.g., Hart et al. 1997; Roy et al. 2015). Of course, other factors—beyond mere quantitative aspects of the parental linguistic input—play an important role in word learning: for instance, lexical category (e.g., nouns vs. verbs) affects patterns in early child lexical development (Gillette et al. 1999), and various qualitative aspects of the input such as responsiveness, joint attention, among others (e.g., Hoff 2006; Rowe et al. 2017) (for a review, see Rowe and Snow 2020; for a metaanalysis of the interplay of quality and quantity of input in child language skills, see Anderson et al. 2021). Here, our point is that all else being equal, one might expect children hearing more necessity modals to master them earlier.

Our study aims at answering two questions. First, is the *Necessity Gap* specific to English learners or does it extend beyond English? Second, are English children's difficulties with necessity modals related to their limited exposure or do they point to deeper difficulties, be they conceptual, semantic or pragmatic? To answer these two questions, we investigate how children learn modals' force in French, a language where the frequency pattern is reversed from English: necessity modals are more frequent than possibility modals in parental speech. We present results from a detailed corpus analysis of 2- to 6-year old French children's spontaneous modal productions and their input, completed by two experimental studies based on the corpus, using the same methods as Dieuleveut et al. (2022) for English to allow a straightforward comparison between the two languages. Our results show that, despite a higher quantitative exposure to necessity modals, French children manifest the same difficulties as their English counterparts: they start using necessity modals later on, use them less frequently, and when they do, they do not use them in an adult-like way; they tend to "overuse" them, that is, they use them in situations where adults find possibility modals more appropriate. We conclude that the delay found for English cannot be due to low quantity of exposure and must reflect a deeper issue.

The remainder of the paper is organized as follows. Section 2 provides more detailed background: Section 2.1 describes the learning problem (what makes it challenging for learners to determine a modal's force); Section 2.2 reviews the empirical evidence grounding the claim that English-speaking children struggle with necessity modals and the hypotheses put forward to explain their struggles; Section 2.3 outlines the main characteristics of the French modal system and reviews existing acquisition corpus studies on French to situate the novelty of our work. We then turn to our empirical contributions. In Section 3, we present Study 1, which provides quantitative results regarding modals produced by children and their parents, systematically comparing French to Dieuleveut et al.'s (2022) results on English. In Sections 4 and 5, we present two experimental studies based on the corpus. Study 2 (Section 4) adapts Dieuleveut et al.'s (2022) paradigm to assess the extent to which children's use of modals is adult-like, that is, testing whether children use their modals in appropriate conversational contexts as judged by adult participants. We replicate Dieuleveut et al.'s (2022) findings in French; Additionally, we extend the experiment to test older children's (4- to 5-year-olds) productions, where we find that difficulties persist. The second experimental study (Study 3, Section 5) is a follow-up carried out to address one potential explanation of results from Study 2; namely, that lower performance on children's necessity modals would stem from adult participants' expectations that children use possibility modals more. Results show no evidence that speakers' attributed age would affect adult participants' choices. In Section 6, we discuss implications of our findings and possible hypotheses for explaining children's misuses of necessity modals, and outline directions for future research. Section 7 summarizes the conclusions gained from this research.

2 Background

2.1 The learning problem

Taking the perspective of the learner, there are three main reasons why figuring out the force of modals—the correspondence between words such as *can* and POSSIBILITY on the one hand, and *must* and NECESSITY on the other hand—is a challenging task: a logical "subset problem," an "observation problem," and a "flavor-flexibility problem."

First, as put forward in the introduction, the entailment relationship such that necessity modal statements (1b) entail possibility modal statements (1a) creates a logical *subset problem* for learners (Berwick 1985; Manzini and Wexler 1987; see Dieuleveut 2021; Dieuleveut et al. 2022, for earlier discussion of the case of modals).² If we take a simplified model of word meaning acquisition, and assume that the learner's hypothesis space for a novel modal's meaning consists of POSSIBILITY and NECESSITY,

²The *subset problem*, also sometimes called *entailment problem*, was originally discussed for syntax, and then for semantics, where it holds both at the level of word meanings (e.g., concrete nouns such as *dog/animal*, see Piantadosi 2011; Xu and Joshua 2007; quantificational determiners such as *some/every*, see Rasin and Aravind 2020) and for scopally ambiguous sentences (see, e.g., Crain and Thornton 1998; Gualmini and Schwarz 2009). The problem takes its name from the fact that the meaning (understood as extension) of a word or sentence is a subset of the other's (see Figure 1).

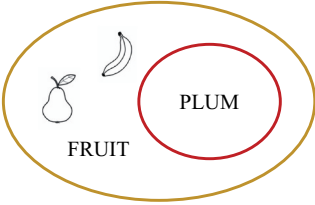
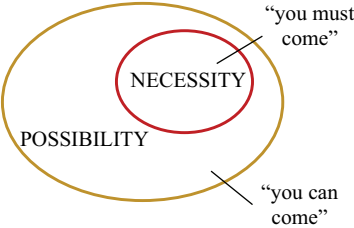
(i) Content words (<i>fruit</i> vs <i>plum</i>)	(ii) Modals (<i>can</i> vs <i>must</i>)
	
The set of plums is a subset of fruits.	The set of necessity situations is a subset of possibility situations.
All plums are fruits.	What is necessary is also possible.
“this is a plum” entails “this is a fruit”	“you must P” entails “you can P”
Taking the learner’s perspective	
<i>Trying to figure out the meaning of a new noun:</i>	<i>Trying to figure out the force of a new modal:</i>
“This is a sig”	“You sig come”
• If I assume <i>sig</i> means PLUM, but in fact <i>sig</i> means FRUIT, my hypothesis can be falsified: I will hear <i>sig</i> used to refer to other fruits (e.g., pears), where its use would be inappropriate if it just meant PLUM. • If I assume <i>sig</i> means FRUIT, but in fact <i>sig</i> means PLUM, my hypothesis will remain consistent: all plums are fruit, so no contradiction arises. My hypothesis remains logically compatible with every context in which plums are mentioned.	• If I assume <i>sig</i> means NECESSARY, but in fact <i>sig</i> means POSSIBLE, my hypothesis can be falsified: I will hear <i>sig</i> used in situations of mere possibility, where its use would be inappropriate if it meant NECESSARY. • If I assume <i>sig</i> means POSSIBLE, but in fact <i>sig</i> means NECESSARY, my hypothesis will remain consistent: Situations of necessity are also situations of possibility, so no contradiction arises.

Figure 1. Set/Subset entailment relationships and resulting acquisition *subset problems* in the domains of (i) content words and (ii) modal words. Note, for clarification, that with modals the logical entailment holds only within a type (“flavor”) of modality. For instance, it is true that if one is obliged to wear a helmet, then one is also permitted to wear it, or likewise, that if it is certain that it is raining, then it is also epistemically possible; but one may be required to come without being physically able to do so.

a problem arises for necessity modals. If the child (incorrectly) assumes that a necessity modal (e.g., *must*) means POSSIBILITY, since there is no situation that makes a necessity modal statement true (e.g., (1b), “you must come”) while making a corresponding possibility modal (1a) (“you can come”) false, they have no incentive to revise their incorrect hypothesis.³ Conversely, if the child (incorrectly) assumes that a possibility modal (e.g., *can*) expresses NECESSITY, they may have opportunities to rectify it, since they may hear it used in non-necessity situations. Therefore, what prevents children from thinking that necessity modals simply express POSSIBILITY? Figure 1 summarizes the *subset problem* and highlights the parallel between modals and content words. Likewise, if children (incorrectly) assume that a word such as *plum* means FRUIT, what kind of evidence can they use to revise this hypothesis? They will only see *plums* used in situations that are compatible with their hypothesis. Whereas conversely, if they incorrectly assume that *fruit* means PLUM, they can get rid of their wrong hypothesis, by hearing *fruit* used to refer to an object that is not a plum.

The second aspect that complicates the learning task is that—unlike what happens with concrete nouns—it is not clear what children can “observe.” Modals such as *must* have quite abstract meanings: they introduce “displacement,” that is, they allow us to refer to some events and situations that are not

³At least, in Upward-Entailing environments. In negative (Downward-Entailing) environments, the direction of logical entailments is reversed: not possible entails not necessary. See Gualmini and Schwarz (2009), who argue that this is a “logical” way to solve *subset problems*, and Dieuleveut (2021) for further discussion.

actual. This *observation problem* has also been famously discussed for attitude verbs such as *think* or *want* that refer to mental states, and as such, lack clear reliable physical correlates in the environment (see Gleitman's et al. 2005 “hard words”; for overviews of the very rich literature on the topic, see, e.g., Hacquard and Lidz 2019; Hacquard and Lidz 2022; and references therein). But modals introduce even a further layer of complexity, coming from their “flavor flexibility.” While attitude verbs always and consistently refer to the same type of modality (e.g., *think/know* always relate to knowledge, *want* to desires), modals auxiliaries display a peculiar kind of ambiguity. That is, they can express different types of possibility or necessity, traditionally called “flavors” in the semantic literature (see Kratzer 1977, 1991, a.o.).⁴ For instance, *devoir* “must” can express a necessity given what is known (e.g., (2a), “it is *certain* that Tom is sleeping”) (traditionally called *epistemic* modality), but it can also—in a different context—express necessity given some rules ((2b), “Tom is *required* to sleep,” traditionally called *deontic* modality), or a necessity given Tom's goals ((2c), “to be in good shape, Tom *should* sleep,” *teleological* modality), among others. Typologies of modal sub-flavors vary; it is common to draw a major split between epistemic and “root” flavors, which subsume all non-epistemics (Hofmann 1976; Kratzer 1977; see Hacquard 2011 for an overview).

- | | | | |
|-----|--|--------------|-----------|
| (2) | Tom | doit | dormir. |
| | Tom | must.3SG.PRS | sleep.INF |
| | 'Tom must sleep/Tom must be sleeping' | | |
| | a. Epistemic: according to what we know, it is certain that Tom sleeps | | |
| | b. Deontic (root): according to the rules, Tom is required to sleep | | |
| | c. Teleological (root): to achieve his goals, it is necessary for Tom to sleep | | |
| | d. Ability (root): according to his physical capacities, Tom is compelled to sleep | | |

In languages such as French and English, modal auxiliaries, as well as some adjectives (e.g., *possible*) and adverbs (e.g., *forcément* “necessarily”), vary in flavor while lexically encoding force: *doit* in (2) always expresses necessity, but can take many flavors. Children have to acquire these two dimensions, force and flavor, in tandem.⁵

2.2 What do we know about the acquisition of modal force?

Prior studies on modal acquisition have used different methodologies to assess children's mastery of modals' force.⁶ In this section, we first summarize results from comprehension studies, and then, from production studies.

Experiments testing children's comprehension of modal sentences have led to two important results. The first one is that 4- to 5-year-old children tend to judge sentences expressing simple possibility as true in necessity situations—for instance, when it is certain that there is a bear in the box, children accept sentences such as “There **might** be a bear in the box” (Noveck 2001; Ozturk and Papafragou 2015; Moscati et al. 2017; Leahy and Zalnieriunas 2021). The second result, more puzzling from the adult stance, is that children also tend to over-accept sentences with necessity modals in possibility situations—for instance, when there are several possibilities open regarding the location of the bear, they tend to accept sentences such as “There **has**

⁴The nature of this flavor “flexibility” is a matter of debate (for review see, e.g., Hacquard 2011). Here we use the term ambiguity descriptively; we remain neutral with respect to the theoretical semantic analysis.

⁵While we leave aside these aspects, the problem is further deepened by the irregularities of scope interactions between modals and negation: in many languages, necessity modals tend to outscope negation (see Iatridou and Zeijlstra 2013; we briefly come back to this in Section 2.3); as well as pragmatic “polite” uses of modals: possibility modals are used to soften statements in a polite way (e.g., “You could be a little more quiet” is commonly used to perform an order to be quiet, closer to necessity; see Grice 1975; Searle 1975). They are substantially discussed by Dieuleveut (2021), Dieuleveut et al. (2022).

⁶Here we do not aim for an exhaustive review of the literature on modal language acquisition, which would have to include much more diverse ways to express modality: mental verbs, *if* conditionals and counterfactuals, mood distinctions, etc. Here we focus on modal auxiliaries/semi-auxiliaries, as this is where the *Necessity Gap* was identified.

to be a bear in the box” (Ozturk and Papafragou 2015; Moscati et al. 2017; Staniszewski et al. 2023; Bleotu et al. 2025 for the adverbs *poate* “maybe” versus *sigur* “certainly” in Romanian). While many studies focus on epistemic modals, making one plausible explanation for children’s behavior coming from flavor variability, this finding has been replicated with root modals. Likewise, 4- to 5-year-olds tend to accept sentences such as “to go to the library, Tom **has to** go down the blue road” in scenarios where two different roads allow one to reach the library (Cournane et al. 2025; Lungu et al. 2026).

Children’s non-adult-like behaviors in such comprehension tasks have been explained in different ways. The first result (children’s over-acceptance of possibility modals in necessity situations) has substantially been discussed in the context of children’s acquisition of scalar implicatures (i.e., inferences one can draw from a logically weaker statement, such as “Tom ate **some** of the plums,” that the logically stronger statement does not hold—i.e., he did not eat them all; see Horn 1972; Grice 1975). Possibility modal statements such as “There **might** be a bear in the box” commonly triggers, among adults, the inference that it is not *certain* that the bear is in the box (since otherwise, the speaker would have chosen a more informative statement, “There **has to**”). Children, when asked to evaluate these underinformative sentences, more readily accept them as true, making them appear “more logical” than adults (Noveck 2001). Although here we will not review in detail the explanations that have been entertained for children’s behaviors (for reviews of the literature on children’s “difficulties” with scalar implicatures, see, e.g., Katsos 2014; Noveck 2018), this literature has shown that children’s propensity to make scalar implicatures depends notably on their ability to access lexical (stronger) alternatives (Chierchia et al. 2001; Barner et al. 2011; Skordos and Papafragou 2016).

The second result, children’s over-acceptance of necessity modals in possibility situations, is a priori more puzzling from an adult perspective. Indeed, while possibility modals are logically true in necessity situations (and under some circumstances, adults also use and accept them), this is not the case for necessity modals: from a logical standpoint, it is false to say “you **have to** go down the blue road” if several distinct roads allow you to reach your goal (on the relevant modal flavor). One hypothesis proposed to explain why children accept such sentences is that children experience processing or conceptual difficulties when reasoning in situations that involve uncertainty (Noveck et al. 1996; Ozturk and Papafragou 2015; Moscati et al. 2017). When faced with situations involving multiple possible outcomes—for instance, a toy hidden in a box—children would tend, even in the lack of decisive evidence, to commit to a conclusion. This tendency, known as *premature closure* (Piérait-Le Bonniec 1987; Acredolo and Horobin 1987), leads them to select one possibility while disregarding alternatives. According to a second line of explanation, children’s difficulties lie at a more fundamental conceptual level, namely in their representations of possibilities (Leahy and Carey 2020; Leahy and Zalnieriunas 2021; Leahy 2024). According to Leahy and Carey (2020), children initially rely on “minimal representations” of possibilities, which differ from adult-like modal representation in that they consist of simple simulations of possible outcomes, and lack logical structure (whereas the mature representational system has symbols that mark representations as necessary, possible, or impossible). On this view, children gradually construct full modal representations through exposure to and use of language. Leahy and Zalnieriunas (2021) argue that even 5-year-olds still lack full-fledged modal representations; as a result, they shall fail to distinguish *can* from *have to* (for arguments against this view, see, e.g., Cesana-Arlotti et al. 2018; Turan-Küçük and Kibbe 2024).

Production studies have examined children as young as two—the age at which children typically begin to produce modals—and have yielded two important complementary findings. First corpus studies of children natural productions have mainly focused on the flavor dimension (for overviews, see Papafragou 1998; Cournane 2020). They have shown that children tend to use modals for root flavors first (i.e., to express abilities, obligations and goals), while they use them for epistemics (i.e., possibilities and necessities given what is known) on average only one year later, from around age three. This *Epistemic Gap*, a term introduced by Cournane (2015), is found consistently across languages (for English, see Kuczaj and Maratsos 1975; van Dooren et al. 2022; for Greek: Stephany 1983; for Korean; Choi 2011; for French:

Bassano 1996; Cournane and Tailleir 2020; for Dutch:; Van Dooren et al. 2019; for Serbo-Croatian, Veselinović and Cournane 2020; for Romanian: Avram and Gaidargi 2021). Corpus studies explicitly focusing on force are rarer and have focused only on English. Dieuleveut et al. (2019, 2022) provide a detailed assessment of the modals English children produce between age two and three based on the Manchester corpus (Theakston et al. 2001) ($n = 12$ children), using a combination of corpus analyses and experiments. Their results reveal an asymmetry in children's mastery of possibility and necessity modals: while children use possibility modals such as *can/might* early on, frequently, productively (i.e., with/without negation, in declarative and interrogative sentences), and use them in an adult-like way, they start using necessity modals such as *have-to* later on, use them less often and in less diverse syntactic environments, and use them in a non-adult-like way: they sometimes use them when adults expect possibility modals. To get an assessment of children's uses, Dieuleveut et al. have developed a variant of the *Human Simulation Paradigm* (Gillette et al. 1999)—a paradigm we will reuse in the current study. Their paradigm allows one to assess how adult-like children's modal productions were, by asking adult participants to guess the force of modals uttered by children and their mothers in snippets of real conversations (dialogs extracted from the corpus) in which the modals were blanked out. The accuracy in determining the modal originally produced by the child is used as a measure of how adult-like children's uses are, as it reflects whether children used their modals in contexts where adults do the same. Their experiment's results show that adult participants' choices align with 2- to 3-year-olds use for possibility modals—that is, children use possibility modals in an adult-like way. However, performance is lower (i.e., agreement rates are lower) on children's necessity modals, which means that children sometimes use necessity modals in situations where adults use possibility modals (e.g., when adults expect “I can see a cat,” children say “I have to see a cat”; Dieuleveut et al. 2022, 13). By contrast, their baseline condition—where the same experiment was run on parental productions—shows no difference in accuracy between possibility and necessity contexts. Based on these results, Dieuleveut et al. (2022) (see also Dieuleveut 2021; Cournane et al. 2025) developed another hypothesis for children's struggles with necessity modals, according to which the core of the problem is not at the conceptual level, but rather in the process of learning word meanings itself—the challenge of mapping words to their meanings. The hypothesis is that children have not yet figured out the right underlying force of their necessity modals.⁷ As they argue, this accounts for results in both production and comprehension: If children do not know *must* expresses NECESSITY, they will over-accept possibility modals in necessity situations (i.e., not make scalar implicatures), since under any account of scalar implicatures, knowing the elements of a scale is a prerequisite for accessing them.⁸ And if children think that *must* expresses POSSIBILITY, it is unsurprising that they accept *must*-statements in possibility situations.

To summarize, findings from comprehension and production studies on English converge in showing a *Necessity Gap* (Dieuleveut 2021): children master possibility modals early, but struggle with necessity modals, treating them in a non-adult-like way. However, the source of children's difficulties with necessity modals is unclear. While various hypotheses have been proposed, a major issue is that the debate only draws on data from English, and the current empirical literature offers no insight into the cross-linguistic generality of the phenomenon. This focus on English is problematic, given that in English, necessity modals are much less frequent than possibility modals in parental speech (Dieuleveut et al. 2022). The English input asymmetry could drive the observed delay in acquiring necessity modals; at the very least, this confound obscures the picture by limiting our ability to evaluate the proposed aforementioned hypotheses. Would English children master necessity modals earlier, if they heard them used more often? This question motivates the current study of French, where we will find that necessity modals are more frequent than possibility modals in parental speech, allowing us to test directly for the role of input frequency in modal force acquisition. We now turn to a brief overview of the French modal system.

⁷There is a weaker and a stronger version of this hypothesis: either children would be uncertain (underspecified meaning), or they have lexicalized necessity modals as possibility modals, potentially reflecting a general bias toward possibility meanings (Cournane et al. 2025).

⁸As noted above, various studies show that children's as well as adults' propensity to make scalar implicatures relies on their ability to access lexical alternatives (Chierchia et al. 2001; Barner et al. 2011; Skordos and Papafragou 2016). If children do not yet master the meaning of logically stronger scalar terms, they will not be able to derive the inference—as if they did not have modal scales in their language (see Deal 2011).

2.3 The French modal system

The French modal system shares many properties with English. In French, *pouvoir* (see (1a), repeated in (3)) is the only modal auxiliary expressing POSSIBILITY; *devoir* (see (1b), repeated in (4)), *falloir* (5), and *avoir à* (6) express NECESSITY (see Chu 2008, for overviews). Syntactically, French modals are typically treated as semi-auxiliaries (as opposed to *vouloir* “want” or *savoir* “know”, which are main verbs) (see Benveniste 1965; Hacquard 2006; Borgonovo and Cummins 2007; Veselinović 2019; Dieuleveut and van Dooren 2025). As in English, French modals allow both root and epistemic flavor interpretations (illustrated in (2)). Unlike in English, French modals inflect for tense, mood, aspect, and agree with their subject. They prototypically embed infinitival complements (see (3), (4), (5a), (6)). Note that the necessity modal *falloir* is special in that it only appears in impersonal constructions (with the expletive subject *il* “it”): Thus, (5c) is ungrammatical. *Falloir* can also take finite (CP) complements, introduced by *que* “that” (see (5b)).⁹ *Avoir à* “have to” is accordingly rare and belongs to a higher register.

(3)	Tu	peux		venir.		POSSIBILITY	
	You	can.2SG.PRS		come.INF			
	'You can come'						
(4)	Tu	dois		venir.		NECESSITY	
	You	must.2SG.PRS		come.INF			
	'You must come'						
(5)	a.	Il	faut		venir.		NECESSITY
		EXPL	must.3SG.PRS		come.INF		
	'It is necessary to come'						
	b.	Il	faut		que tu		viennes.
		EXPL	must.3SG.PRS		that you		
	'It is necessary that you come/You should come'						
	c.	*Tu	faux		venir.		
		*You	must.2SG.PRS		come.INF		
	(intended): 'You should come'						
(6)	Tu	as	à		venir		NECESSITY
	You	have.2SG.PRS	to		come.INF		
	'You have to come'						

There are few corpus-based studies on modal acquisition in French. Based on the analysis of the naturalistic productions of one child between 1;9 and 4;0, focused on the emergence of epistemic talk, Bassano (1996) reports that modal auxiliaries are used for root flavors before epistemic ones (in line with the *Epistemic Gap*, Cournane 2015). Note that because Bassano's goal was to study the early development of epistemic talk, her study extends to a broader range of expressions conveying epistemic notions, including attitude verbs such as *penser* “think,” adjectives such as *possible*, adverbs such as *peut-être* “maybe,” futurate *aller* + infinitif “be going to,” hypothetical and frozen expressions such as *je sais pas* “I don't know.” More recently, while also focusing on the emergence of epistemic talk, Cournane and Tailleur (2020) provide French data from a larger corpus (based on the Paris Corpus; $n = 6$ children, Morgenstern and Parisse 2012), where they include the modals *pouvoir*, *devoir*, *falloir*, and *peut-être*. Notably, they report ages of first production of *pouvoir* and *falloir* (*pouvoir*: between 1;07 and 2;03; *falloir*: between 1;10 and 2;09; first occurrences of *devoir* appear later (from 2;01 to 4;05). Neither Bassano's nor Cournane and Tailleur's studies include data regarding the input. Finally, a third study from Jeretić (2018), also based on the Paris corpus, but focusing on the

⁹The embedded verb takes the subjunctive. *Pouvoir* and *devoir* also admits impersonal uses and CP complements, when adding the reflexive clitic *se* (e.g., “*Il se peut que Tom dorme*” “It may be that Tom sleeps”), but these uses are rare in spontaneous spoken interaction.

acquisition of root modals (i.e., excluding epistemic uses) and their interaction with negation, mentions that French children produce slightly fewer necessity modals as compared to adults (50% of all children's modal utterances included in her study express necessity, vs. 58% for adults') (modals included in Jeretič's 2018 study: possibility: *pouvoir*, *avoir le droit* "have the right to"; necessity: *falloir*, *devoir*, *avoir besoin* "have the need to," *être obligé* "be required to").¹⁰

While informative, these studies only offer a partial view of the issue. None of them allows a direct comparison with Dieuleveut et al.'s (2022) study, since they focus on either epistemic uses or root modals, and include other ways to encode modality. Additionally, all of these studies omit the necessity modal *avoir à*, which although infrequent, is still used. Our study aimed to enable a more systematic and direct comparison between French and English, by using the same methods and inclusion criteria as Dieuleveut et al. (2022). In our study, we thus focused on French modal semi-auxiliaries (possibility: *pouvoir*; necessity: *devoir*, *falloir* and *avoir à*; modals included in the English study: possibility: *can*, *could*, *might*, *may*, *able*; necessity: *must*, *should*, *need (to)*, *have to*, *got to*, *supposed to*, *ought to*). First, we assess when children start using modals in French and how frequent possibility and necessity root and epistemic modal uses are in their production, and how this evolves from 1 to 6 years of age, using a larger sample than in prior studies on French ($n = 11$ children). Second, we provide a detailed assessment of French modal input (parental speech). Third, we assess how adult-like children's spontaneous modal uses are, by testing whether they use them in the appropriate conversational contexts, building on the experimental paradigm introduced by Dieuleveut et al. (2022).

3 Study 1: Corpus study of children's and adults' modal productions

3.1 Materials and methods

3.1.1 Corpora selection

We used the Lyon Corpus of French (Demuth and Tremblay 2008) (5 child-mother pairs; age range: 1;00–3;00; 3 females, 2 males) and the Paris corpus (Morgenstern and Parris 2012) (6 child-mother pairs; 3F, 3 M; age range: 0;07–6;03) from the CHILDES database (MacWhinney 2000).¹¹ For English, we borrow data from Dieuleveut et al.'s (2022) study, which they base on the Manchester corpus of UK English (Theakston et al. 2001) (CHILDES, MacWhinney 2000) (12 child-mother pairs; 6F, 6 M; age range: 1;09–3;03; included age range: 2;00–3;03). Children were recorded in unstructured play sessions during interactions with their parents at home. Both French corpora contain audio/video data, which we used in particular to disambiguate flavor.

3.1.2 Exclusion criteria

For French, we focused on modal semi-auxiliaries (Possibility: *pouvoir*; Necessity: *devoir*, *falloir*, *avoir à*), to make our data directly comparable to Dieuleveut et al.'s (2022) study. We therefore exclude other means to encode modality: adverbs such as *peut-être* "maybe," adjectives such as possible, verbs such as *penser* "think" or *vouloir* "want," as well as modal verbal inflection (imperative, conditional, subjunctive).¹² We also exclude the semi-auxiliary *aller* ("go"), as it primarily expresses futurity.¹³

¹⁰Jeretič also reports data about Spanish ($n = 6$ children), which suggests that in Spanish as well, necessity modals are slightly more frequent than possibility modals in the input (53% of all adult modal utterances, vs. 48% for children modal productions) (modals she includes for Spanish: possibility: *poder* "can," *tener derecho* "have the right"; necessity: *tener que*, *hay que*, *deber*, *necesitar*). Note that Jeretič's study on French only includes 5 of the 6 children from the Paris corpus.

¹¹At the time when the extraction was run, the Paris corpus only included 6 children. Note that it now has data from one supplementary child, Philippine (from age 4;08;27 to 5;11;11). Analyses only include transcripts that were available at the time.

¹²Modals included in Bassano's (1996) study: modal verbs, adjectives, adverbs, *aller* + inf, hypothetical, expressions such as *je sais pas*. In Courneau and Tailleux's (2020) study: *pouvoir*, *devoir*, *falloir*; *peut-être*. In Jeretič's (2018) study: *pouvoir*, *avoir le droit*, *falloir*, *devoir*, *avoir besoin*, *être obligé*. None of these studies includes the semi-auxiliary *avoir-à* ("have-to").

¹³There are debates in the semantic literature about the status of "will/shall/be going to", and whether they should be analyzed as modals or tenses conveying a purely future temporal meaning (e.g., see Copley 2009). Here we followed Dieuleveut et al.'s (2022) choice to exclude "will/would/shall/be going to" from their study.

3.1.3 Extraction and coding

We extracted all utterances containing modal semi-auxiliaries (total number of utterances in Fr(ench): adult: 5,231; 2–3-year-olds: 1,514; 3–5-year-olds: 1,404; excluding repetitions:¹⁴ adult (2.2%): 5,114; 2–3-year-olds: (8.9%): 1,379; 3–5-year-olds: 1,296 (7.7%); in En(lish): adult: 20,755 utterances; child: 5,842; excluding repetitions and tag-questions: adult (9.1%): 18,853; child (17.8%): 4,800). We coded all utterances for force (possibility vs. necessity), as shown in (7), flavor (epistemic vs. root),¹⁵ as shown in (8), complement, as shown in (9), negation, as shown in (10), and person subject. Data were coded by the first author. To code for flavor, we relied on contextual information provided in the CHILDES transcripts.¹⁶ To assess coding reliability, a second coder (native speaker of French) independently coded 300 utterances for each category. Inter-coder reliability was calculated using Cohen's Kappa (Landis and Gary 1977). The resulting Kappa values were: for force, $\kappa = 1$; for flavor, $\kappa = 0.89$; for complement, $\kappa = 0.95$; for negative environments, $\kappa = 0.96$. The data, a transcription guide, and double codes are available at <https://osf.io/xqn8b/overview>.

(7) Modals by Force:

- a. Possibility: Fr: *pouvoir*; En: *can, could, might, may, able to*
- b. Necessity: Fr: *falloir, devoir, avoir à*; En: *must, should, need (to), have to, got to, supposed to, ought to*

(8) Flavor:

- a. root:
MAMAN: y a plein d'habits sales! ('There are many dirty clothes!')
- MAMAN: Elle **doit** laver tout le linge ('She **must** do all the laundry')
- b. epistemic:
ENFANT: je trouve pas la grosse ('I can't find the big one')
- MAMAN: elle **doit** être restée dans la voiture ('It **must** have stayed in the car')

(9) Complement:

- a. + NP (excluded): 'Il **faut** [du pain]' ('We need [bread]')
- b. + Inf: 'Il **faut** [venir]' ('it is necessary [to come]')
- c. + Finite: 'Il **faut** [que tu viennes]' ('it is necessary [that you come]')

(10) Negative environments:

- a. No negation: 'Il **peut** venir' ('He can come')
- b. Negation:
- on modal: 'Il (ne)¹⁷ **peut** {pas/plus . . . } venir' ('He **can't/cannot** come')
- on higher auxiliary/embedding verb: 'Je (ne) pense pas que tu **aies à** le chercher' ('I don't think you **have to** look for it')
- other negative (downward-entailing) environments: 'Personne (ne) **doit** partir' ('Nobody must go')

¹⁴We consider as repetitions cases where the speaker (child or adult) repeats a sentence uttered right before by herself or by another speaker.

¹⁵We did not differentiate among various root subflavors (e.g., ability, teleological, deontic) because most cases were too ambiguous. For insightful discussion of the methodological challenges raised by corpus-based modality research, see Reisinger et al. (2022). They note that quite generally, corpus-based studies on modal flavor find high coder consistency for highest-level distinctions (root vs. epistemic), but a lack of reliable inter-annotator agreement on fine-grained distinctions (e.g., Rubinstein et al. 2013).

¹⁶Most uses were straightforward to code based on contextual cues (e.g., clear expression of rules, goals, or abilities, as illustrated in (8a), as opposed to (un)certainity about events, as illustrated in (8b)). The few problematic cases involved *devoir*+BE and cases of *pouvoir* in the conditional, but represent very few of the data (6.7% of uses of *pouvoir* are with conditional, i.e., 2.3% of the whole adult data). As a diagnostic to code for flavor, we replaced the sentence by paraphrases being unambiguously root (e.g., "it is allowed/required that p") or epistemic ("it is probable/likely that p"). Note, interestingly, that for cases involving *devoir*+BE, prosodic information sometimes helped determining flavor: we observed that when focus is placed on the modal itself (e.g., "Il DOIT être dessous" "It MUST be under"), the epistemic interpretation seems unavailable, whereas when focus is on the complement (e.g., "Il DOIT être dessous" "It MUST be under"), the interpretation is more naturally epistemic. A similar observation was made for English by Havron et al. (2023). We intend to investigate further those facts in the future.

¹⁷French has two discontinuous negative markers, *ne* and *pas*, that jointly express negation. *Pas* is the main negator; *ne* can be left out. Note that necessity modals *falloir*, and arguably *devoir*, take wide scope with respect to negation: (ii) means "necessary not P" but not "not necessary that P." By contrast, *pouvoir* always take narrow scope ((i) mean "not possible" but not "possible not P"). The availability of the "not necessary" interpretation for *devoir* (iii) is debated, and might require special focus on *devoir* (see Homer 2015, Jeretič 2018). *Avoir à* (iv), just as English *have to*, scopes under negation.

(i) Tu ne peux pas venir	('You can't come')	NOT > POSS	*POSS > NOT
(ii) Il (ne) faut pas venir	('You mustn't come')	*NOT > NEC	NEC > NOT
(iii) Tu (ne) dois pas venir	('You mustn't come')	?NOT > NEC	NEC > NOT
(iv) Tu (n') as pas à venir	('You don't have to come')	NOT > NEC	*NEC > NOT.

3.2 Results

Overall, utterances containing modal semi-auxiliaries represent 3.9% of all adult utterances in French (vs. 5.8% in English), and 1.9% of child utterances between age 2 and 3 (vs. 2.4% in English).¹⁸ Tables 1 and 2 summarize counts of adult and child modal productions by force and age range, with the proportion of epistemic uses and negation, in French (Table 1a,b) as compared to English (Table 2). We sequentially describe the results for each parameter of interest (force, flavor, and negation) in the following sections, first focusing on the input (i.e., parental production) and then on children's productions. We present French and English on a par to ease comparison. Results about subject (not reported by Dieuleveut et al. 2022) are available in the Online Supplemental Appendix A <https://osf.io/xqn8b/overview>.

In adult speech, we find a substantial difference between French and English in the relative frequency of possibility and necessity modals. While in English, possibility modals are more frequent than necessity in the input (Possibility: 72% of all their modal utterances, vs. Necessity: 28%), in French, necessity modals are more frequent than possibility modals (Poss: 38% vs. Nec: 62%). A chi-square test of independence shows that the distribution of modal force (necessity vs. possibility) differs significantly between French and English adults ($\chi^2(1, N = 23,960) = 1833, p < .001$, Cramer's $V = .28$). Upon further inquiry, the difference appears to come from the fact that in English, *can* is extremely frequent: it occurs in 3.3% of all mothers' utterances, twice as much as *pouvoir* (1.6%). Contrastively, the necessity modal *falloir* is particularly frequent in French (2% of all utterances in the corpus; vs. 0.7% for the English most frequent necessity modal, *have to*).²⁰ *Avoir à* is extremely rare and appears almost exclusively in so-called "exceptive" constructions (e.g.,

Table 1. Counts and percentages of modal uses by force and modal in French, for adults and children, by age range (1a: 2- to 3-year-olds; 1b: 3- to 6-year-olds), ordered by modal frequency, with usage frequency with negative markers and frequency of epistemic uses.

	ADULTS (n = 5,107)						CHILDREN, 2–3-year-olds (n = 1,379)					
	Count	Prop.	Count Neg	Prop. Neg	Count epi	Prop. epi	Count	Prop.	Count Neg	Prop. Neg	Count epi	Prop. epi
1a												
POSS	2,008	39.3%	439	21.9%	54	2.7%	850	61.6%	146	17.2%	2	0.2%
NEC	3,099	60.7%	539	17.4%	249	8.0%	529	38.4%	80	15.1%	2	0.4%
<i>falloir</i>	2,650	51.9%	471	17.8%	0	0%	492	35.7%	64	13.0%	0	0%
<i>devoir</i>	403	7.9%	25	6.2%	249	61.8%	21	1.5%	0	0%	2	9.5%
<i>avoir-à</i>	46	0.9%	43	93.5%	0	0%	16	1.2%	16	100.0%	0	0%
	ADULTS (n = 2,543) ¹⁹						CHILDREN, 3–6-year-olds (n = 1,294)					
	Count	Prop.	Count Neg	Prop. Neg	Count epi	Prop. epi	Count	Prop.	Count Neg	Prop. Neg	Count epi	Prop. epi
1b												
POSS	1,111	43.7%	223	20.1%	11	1.0%	741	57.3%	137	18.5%	1	0.1%
NEC	1,432	56.3%	199	13.9%	115	7.9%	553	42.7%	69	12.5%	25	4.5%
<i>falloir</i>	1,146	45.1%	154	13.4%	2	0.1%	402	31.1%	52	12.9%	0	0%
<i>devoir</i>	255	10.0%	17	6.7%	113	44.3%	136	10.5%	5	3.7%	25	18.4%
<i>avoir-à</i>	31	1.2%	28	90.3%	0	0%	15	1.2%	12	80.0%	0	0%

Repetitions and instances taking NP complements are excluded.

¹⁸Total number of utterances was computed using the MLU function of the CLAN program (MacWhinney 2000). Proportions out of all utterances are based on mothers' (MOT) productions only. In the rest of the analysis, all adult speakers are included (MOT; OBS; FAT; GDM; AUN; INV; FAM; ADU; BAB; HOU; FA1; TEE; UNC; VIS; GRM; CAM; GDF; VAL; FRI; OB1; GIR; STU; ADU2; EMI; UNI; GER). For child data, we only include CHI. Data from other children are excluded.

¹⁹The reason we split adult's talk between talk toward 2- to 3-year-olds and talk toward older children (Table 1a vs. 1b) is to make it comparable to the English study, where only 2- to 3-year-olds were assessed (Table 2). Note that we do not see much variation in adult's talk between the two age bins. Also note that in their study, Dieuleveut et al. (2022), includes children up to 3;3, so we set the boundary between age bins at 3;03 and not 3;00.

²⁰To better understand the difference between English and French adult usage, we asked a bilingual (native speaker of French) to translate a sample of 200 *can* utterances from English to French. They kept the possibility modal in 68%, but used simple present in 17% (most cases with perception verbs, e.g., "I can see" as "je vois"); they used modal lexical verbs in 12% (*arriver à* "manage," *essayer* "try," *avoir le droit* "be allowed," *être capable* "be able," *empêcher* "prevent"), and adjectives in 2%. Conversely, we asked a bilingual (native speaker of English) to translate a sample of 200 *falloir* utterances from French to English. In most cases, they kept necessity modals (86%; *have to*: 146, *should*: 21, *need to*: 4, *gotta*: 1); they used negated imperatives in 6% (e.g., "faut pas s'énerver" "don't get upset"); negated possibility modals in 3%; *how/whatto* complements in 2.5% (embedded cases); and adverbs like "properly" in 1.5%.

Table 2. Counts and percentages of modal uses by force and modal in English, for adults and children (2- to 3-year-olds), ordered by modal frequency, with usage frequency with negative markers and frequency of epistemic uses.

	ADULTS (<i>n</i> = 18,853)						CHILDREN, 2–3-year-olds (<i>n</i> = 4,800)					
	Count	Prop. (%)	Count Neg	Prop. Neg (%)	Count epi	Prop. epi (%)	Count	Prop. (%)	Count Neg	Prop. Neg (%)	Count epi	Prop. epi (%)
POSS	13,500	71.6%	2,828	20.9%	1,325	9.8%	3,798	79.1%	1,937	51.0%	93	2.4%
<i>can</i>	10,779	57.2%	2,396	22.2%	37	0.3%	3,620	75.4%	1,881	52.0%	1	0%
<i>could</i>	1,213	6.4%	208	17.1%	117	9.6%	86	1.8%	52	60.5%	7	8.1%
<i>might</i>	1,155	6.1%	39	3.4%	1,155	100.0%	80	1.7%	2	2.5%	80	100.0%
<i>able</i>	315	1.7%	181	57.5%	0	0%	3	0.2%	0	0.0%	5	55.6%
<i>may</i>	38	0.2%	4	10.5%	16	42.1%	9	0.1%	2	66.7%	0	0%
NEC	5,353	28.4%	539	10.1%	341	6.4%	1,002	20.9%	52	5.2%	21	2.1%
<i>have to</i>	2,398	12.7%	108	4.5%	6	0.3%	352	7.3%	7	2.0%	1	0.3%
<i>got to</i>	937	5.0%	11	1.2%	7	0.7%	288	6.0%	5	1.7%	0	0%
<i>should*</i>	696	3.7%	159	22.8%	55	7.9%	21	4.5%	4	6.0%	0	0%
<i>need to</i>	493	2.6%	84	17.0%	0	0%	217	2.4%	13	17.5%	2	15.8%
<i>must*</i>	411	2.2%	65	15.8%	264	64.2%	114	0.4%	20	19.0%	18	9.5%
<i>supposed to</i>	335	1.8%	105	31.3%	9	2.7%	9	0.2%	3	33.3%	0	0%
<i>ought to*</i>	83	0.4%	7	8.4%	0	0%	1	0%	0	0%	0	0%
ALL	18,853	100.0%	3,366	17.9%	1,663	8.8%	4,800	100.0%	1,989	41.4%	114	2.4%

Abbreviation: Prop epi, proportion of epistemic uses.

“Tu n’as qu’à mettre le cheval là-bas,” “you only have to put the horse there”) (40/43 cases of negated *avoir-à*). Turning to children’s production, we find that French and English children produce possibility modals more frequently than necessity modals (Fr, 2- to 3-year-olds: Poss: 62% of all their modal utterances vs. Nec: 38%; En: Poss: 79% vs. Nec: 21%). For the 3- to 6-year-olds, we find a slight increase in proportion for necessity modals (Fr: Poss: 57% vs. Nec: 43%). Results of chi-square tests of independence indicate that French adults and 2- to 3-year-old French children differ significantly in their distribution of modal force, with adults producing a higher proportion of necessity modals than young children (χ^2 (1, $N = 6,486$) = 218.6, $p < .001$, Cramer’s $V = .18$). Comparing adults and 3- to 6-year-old French, we also find a medium effect (χ^2 (1, $N = 6,401$) = 135, $p < .001$, Cramer’s $V = .15$). Comparing between 2- to 3-year-old and 3- to 6-year-old French children, we only find a small effect (χ^2 (1, $N = 2,673$) = 5.1, $p = .024$, Cramer’s $V = .04$).

3.2.1 Age of first production

As reported in English, French children tend to produce possibility modals earlier than necessity modals, with a typical gap of about four months. In English, first uses of *can* are found before age 2, whereas *have to* typically appears around 2;04. Likewise, in French, the average age for first use of *pouvoir* is 1;11, while *falloir* and *devoir* appear at 2;03 and 2;11, respectively. Table 3 presents detailed results by children: for each child, it gives both the first recorded use of possibility and necessity

Table 3. Counts, proportion and first usage of possibility and necessity modals by child in the French corpora ($n = 11$) (2- to 3- year olds).

Corpus, Child	Recording span		Counts		Prop. (P vs. N) (%)	First POSS modal (age)	First NEC modal (age)
	from age	to age ²¹	POSS	NEC			
Lyon Anais	1;09,22	3;00,15	99	22	82%	peut plus 'can't anymore' (1;09,22b)	faut chauffer là maman là 'need to heat it up there, mom, there' (2;06,21b)
Lyon Marie	1;00,02	4;00,05	133	49	73%	peut pas 'can't' (1;06,22b)	oh oui oh oui dois le peser 'oh yes, oh yes, need to weigh it' (2;05,16a)
Lyon Marilyn	1;11,18	2;11,14	40	40	50%	peux goûter là 'can taste it there' (2;01,17a)	faut tourner la page 'need to turn the page' (2;07,21a)
Lyon Nathan	2;06,13	3;00,03	9	6	60%	peux ... ça 'can ... that' (2;06,13a)	il faut 'it's necessary' (2;10,22b)
Lyon Theotime	2;00,27	3;00,03	163	55	75%	il peut pas 'he can't' (2;00,27b)	les petites, faut ranger 'the little ones, need to tidy up' (2;03,24a)
Paris Anae	2;01,23	6;03,08	123	39	76%	ahah je peux pas! 'aha I can't' (1;07,03)	faut enlever le scotch 'need to remove the tape' (2;00,00)
Paris Antoine	1;07,03	5;10,30	46	37	55%	peux couper 'can cut' (2;02,12)	il faut pas 'must not' (2;01,23)
Paris Julie	1;09,16	8;01,07	7	13	35%	peux pas 'can't' (1;09,16)	faut appuyer 'need to press' (1;11,11)
Paris Leonard	2;03,00	3;02,25	19	17	53%	peut les manger comme ça 'can eat them like that' (2;03,00)	il faut pas mettre ça 'mustn't put that' (2;03,26)
Paris Madeleine	1;10,07	6;11,26	163	234	41%	peux manger toute seule? 'can eat by herself?' (1;11,13)	faut fermer 'need to close' (1;10,07)
Paris Theophile	1;10,00	4;11,11	48	17	74%	peux pas 'can't' (1;10,00)	faut ramasser! 'need to pick it up!' (2;09,07)
MEAN			77	48	62%	1;11,24	2;03,22

²¹Analyses only include transcripts that were available at the time. See footnote 5.

modals, and the relative usage proportion of possibility/necessity. We see some variation across children, but the overall pattern is consistent: possibility modals emerge earlier than necessity modals.

3.2.2 Negation

Figure 2 gives a visual summary of the differences between French/English adult/child possibility/necessity modal uses with and without negation. In adult speech, negation is more frequent with possibility (e.g., *can't*) than with necessity modals (e.g., *mustn't*), both in English and French (Fr: Poss: 22% with negation vs. Nec: 17%; En: Poss: 21% vs. Nec: 10%). The French negative marker *ne* is overtly expressed in only 5.0% of all negated modal adult utterances. We find few modals in other downward-entailing environments. Anecdotally, in French, we only find one case where *ne* appears under the modal (“il peut très bien ne pas pleuvoir” “it may well not rain”).

Children use negation—which we can take as initial evidence of productivity (Stromswold 1990)—more frequently with possibility than with necessity modals, both in French and English (Fr, children: Poss: 17%; Nec: 15%; En, children: Poss: 51% vs. Nec: 5%; in French, we find no negated *devoir* between age 2 and 3). We find an interesting difference in children’s usage of negated possibility modals: while English children produce negated possibility modals (*can't*) extremely frequently (51%, vs. adults: 21%), French children produce them slightly less often than adults (17% vs. 22%).

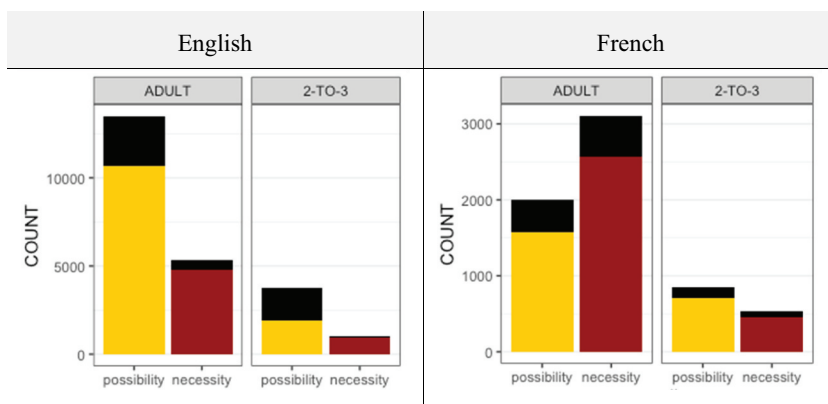


Figure 2. Counts of possibility modal (yellow) and necessity modal utterances (red) with and without negation (black) (En: adult, 18,853 utterances; child, 4,800; Fr: adult, 5,114; child, 1,379). Modals included in English: possibility: *can, could, might, may, able to*; necessity: *must, should, need, have to, got to, supposed to, ought to, need to*; in French: possibility: *pouvoir*; necessity: *falloir, devoir, avoir à*.

3.2.3 Flavor

As reported in English (Dieuleveut et al. 2022; van Dooren et al. 2022), in adult speech, root uses of modals are prevalent as compared to epistemic ones. Epistemic uses are even rarer in French than in English (Fr: 5.9% of all parents’ modal uses; En: 8.8%). We identified only two instances of *falloir*, and no *avoir à*, used epistemically.²² By contrast, *devoir* is more often used for epistemic than root modality (*devoir*: 60% of epistemic uses; similarly, for *must*: 64%). Again, as reported in English, children produce extremely few epistemic modals between the age of two and three (Fr: 0.4%; between age 3 to 6: 1.8%; En: 2.4%). The first child epistemic use we identified in French is at age 2;08 (“ça doit être plein de neige” “it must be full of snow,” Theotime, Lyon corpus, 2;08,25; see also Cournane and Tailleur 2020). Our results thus align with what has been reported for other languages in the literature on the “Epistemic Gap” (see section 2.2). Note that while epistemic uses of modal semi-auxiliaries are rare, we

²²Namely, “FAT: il fallait quand même être un peu tordu” “you had to be a little messed up, honestly,” Paris corpus/Theophile/30410; and “ADU: tu fais de la gymnastique? (...) ADU : faut que tu fasses de la gymnastique là tu dois être super fort !” “Do you do gymnastic? (...) it has to be that you do gymnastics you must be super strong!,” Paris corpus/Theophile/40620.

find uses of epistemic adverbs such as *peut-être* (“maybe”) early on (before 2-years-old for three children), as well as first mental verbs (*savoir* “know” and *connaître* “know” appear before age 2).

3.3 Corpus study: Discussion

French and English input differ in the relative frequency of possibility and necessity modals, with French parents using more necessity modals. Despite this input asymmetry, in both languages, children use necessity modals less frequently than possibility modals, and start using them later on with a “gap” of about 4 months (see Table 3).

First, why this input asymmetry between English and French? A priori, one would expect the relative frequency of necessity and possibility modals to be consistent across languages, yet, French and English differ. Several factors could be involved. Our data show that the difference is mainly driven by the remarkably high frequency of *can* in English: it occurs in 3.3% of all mothers’ utterances in the corpus, whereas *pouvoir* occurs in 1.6% of all French mothers’ utterances. Contrastively, in French, *falloir* is particularly frequent—much more than “have to” (0.7% of all mothers’ utterances). English parents may more routinely use “can” for indirect speech acts (e.g., “You can play now,” “Can you do this?”); also note that an important proportion of *can*-uses involve perception verbs (e.g., “I can see X”), expressions which are less natural in French (Dieuleveut 2023). On the other hand, the high frequency of *falloir* might be related to its special syntactic behavior (see Section 2.3); English parents may rely on alternative ways (such as imperatives) to describe rules or obligations.

Turning to children, several factors might contribute to the prevalence of possibility modals in their early speech. As discussed by Dieuleveut et al. (2022), this may stem from differences between children and adults’ conversational status and goals: children are less often in a position to give orders or express certainty and are therefore less likely to use necessity modals than adults. Moreover, children’s first modals tend to describe abilities (e.g., “I can/can’t do it”) (which lack clear necessity counterparts, Horn 1972), and children may prefer alternative strategies to express orders or to express epistemic certainty, such as using imperatives, or bare sentences. It may also be that children avoid using necessity modals because they would be cognitively costlier, but this hypothesis is hard to assess because existing studies on the processing of possibility and necessity modals remain scarce even with adults. Some researchers (e.g., Jeretič 2018) argue that weak expressions impose greater cognitive demand than strong ones, implying that if processing considerations were involved, the difficulties should be reversed.

Sparse production does not imply lack of comprehension: quantitative data only provide a partial picture of children’s mastery of modals. To get a finer-grained assessment of their uses, we employed the paradigm introduced by Dieuleveut et al. (2019, 2022), which allows to assess whether children use their modals in appropriate conversational contexts.

4 Study 2: An experiment to assess children’s modal uses

4.1 Goals

The key idea of the paradigm is to use adults’ judgments to assess children’s usage in context.²³ Adult participants are asked to guess modals used by children (or their mothers) in dialogs randomly

²³This paradigm, developed by Dieuleveut et al. (2019, 2022), is a variant of the *Human Simulation Paradigm* (Gillette et al. 1999), originally designed to assess the informativeness of various aspects of children’s linguistic input to figure out word meanings, by using adults to simulate word-learning (see White et al. 2017). In the original paradigm, participants have to recover blanked out words in various contexts (e.g., video clip, syntactic frame, associated words); the accuracy with which they recover the original word is taken as a measure of context informativity. The novelty introduced by Dieuleveut et al. (2022, Experiment 5) is to use the paradigm to assess children’s productions. They use the paradigm as a way to evaluate whether children use modals in appropriate conversational contexts (aligning with adult usage). The experiment of mother’s production is turned into a baseline measure of conversational context’s informativity, ensuring that possibility and necessity modals are equally recoverable from conversational context alone. The paradigm has been reused for other cases of word learning, for example, the acquisition of the definite article (*the* vs. *a*) (Ying et al. 2024).

sampled from the corpus (i.e., what modals they would use if they were in the same situation as the child/mother). This allows to test whether children used their modals in appropriate (adult-like) conversational contexts. The accuracy in determining the modal originally produced by the child is used as a measure of how adult-like children's uses were, indicating whether children used their modals in contexts where adults would. High accuracy (i.e., high agreement rate between adults and the child's original use) shows that children have used their modals in an adult-like way. Low accuracy indicates that children did not use their modals in an adult-like way: either they used *pouvoir* in a context where adult participants prefer *devoir/falloir*, or they used *devoir/falloir* in a context where adult participants prefer *pouvoir*. The experiment on mothers' production is used as a baseline for how informative the context is. It demonstrates that force is inferable from conversational context with (we expect) no difference in informativity between possibility and necessity. If children use their modals in an adult-like way, we should have no difference in accuracy between the experiment on child usage and the mothers' baseline (i.e., high agreement rates in both cases). If they do not, we should obtain a lower accuracy for the experiment testing children's utterances than for the baseline.

4.1.1 Testing the effect of input frequency: Predictions

If input frequency matters for child modal mastery, since French children are more exposed to necessity modals in their input as compared to English learners, we should obtain higher performance on French children's necessity modals than what was found for English children, and potentially lower performances on their possibility modals (cf [Tables 1a and 2](#): *falloir*: 52% of all French adult modal utterances; have-to: 13%; *devoir*: 8%; must: 2%; vs. *pouvoir*: 39%, can: 57%).²⁴ Focusing on French, if input frequency matters for child modal mastery, we expect *falloir* leading to higher performance over *devoir*, since *falloir* is much more frequent in the input.

4.2 Methods and materials

4.2.1 Experimental stimuli

Adult participants were presented with various excerpts of conversations randomly sampled from the corpus and asked to fill in the blank in the last utterance, by picking between two options, either a possibility (*pouvoir*) or a necessity modal (*devoir/falloir*). All dialogs consisted of seven utterances and the target modal utterance with a blank, with the two options displayed at the bottom of the screen. [Figure 3i](#) illustrates a trial, for a (necessity) modal originally uttered by a child. [Figure 3ii](#) illustrates a trial for a (necessity) modal originally uttered by a mother.²⁵

4.2.2 Conditions

We tested productions from three age groups (between subject): 2-to 3-year-olds, 4-to 5-year-olds, mothers (used as baseline).²⁶ For each age group, we ran two versions of the experiment varying the modal contrast (between subject): in Study 2a, contexts were extracted and participants had to pick between *devoir* and *pouvoir*; in Study 2b, we contrasted *falloir* and *pouvoir*. We did not test *avoir à* because it is too rare in the corpus (see [Table 1](#)). Force (possibility vs. necessity contexts) was tested

²⁴Dieuleveut et al. (2022) tested both the contrasts between "can/must" and "can/have-to" (which they call, respectively, conditions root-AFF-1 and root-AFF-2). They report 85.1% accuracy on children possibility utterances in root-AFF-1, 79.6% in root-AFF-2, versus 42.6% accuracy on children necessity utterances in root-AFF-1 and 60.2% in root-AFF-2.

²⁵As a clarification, note that we call "possibility contexts" dialogue contexts where the speaker originally decided to use a possibility modal, and "necessity contexts," dialogue contexts where the speaker originally decided to use a necessity modal. This substantially differs from how comprehension tasks—like the ones reviewed in [Section 2.2](#)—define possibility/necessity scenarios. In comprehension tasks, experimenters directly manipulate the situations aiming toward what is most clear about the force of the situations described (e.g., with clearly defined options involving open/closed boxes/roads). With this corpus-based experimental paradigm, one cannot define possibility/necessity contexts in an objective manner (e.g., based on the coders' own evaluations of the conversations). This is also why it is absolutely crucial to use the experiment on mothers' production as a baseline measure of informativity of the context. If adults are at chance for the experiment on mother's utterance, the paradigm cannot be used, since this would mean that context is not informative about force.

²⁶Note that we ran the experiment on 4- to 5-year-olds productions after Study 3 ([Section 5](#)).

(i) CHILD productions	(ii) MOTHER productions
ENFANT : mh. MAMAN : et oui. MAMAN : c'est magique. ENFANT : ah ça revient pas. MAMAN : ... (<i>rit</i>) ENFANT : revient. ENFANT : revient pour mon bébé. ENFANT : euh _____ dormir dans mes bras. doit peut	MAMAN : ah oui. ENFANT : tu as vu comme la maman de Léa dans le train il... il a grossi. ENFANT : il a mangé beaucoup de glaces beaucoup de bonbons. MAMAN : ah oui sûrement. ENFANT : mh. ENFANT : c'est quoi ça ? ENFANT : tu me dis c'est quoi ? MAMAN : ah c'est tu me poses des questions et je _____ répondre. dois peux
CHILD: mh. MUM: yes, indeed. MUM: it's magic. CHILD: ah, it's not coming back. MUM: ... (laughs) CHILD: come back. CHILD: come back for my baby. CHILD: uh _____ sleep in my arms. has to can	MUM: oh right. CHILD: did you see how Léa's mom on the train he... he got fat. CHILD: he ate lots of ice cream lots of candy. MUM: oh yes surely. CHILD: mh. CHILD: what is that? CHILD: tell me what is it? MUM: oh it's you ask me questions and I __ answer. have to can

Figure 3. Example trials, necessity contexts, Study 2a, with English translations. (i) Experiment on children's production. (ii) Experiment on mothers' production (used as baseline).

within subject. Note that contra Dieuleveut et al. (2022), we only tested root modals because epistemic uses are too scarce in French children's production (see Table 1a); We did not test negated utterances because of the fact that both *falloir* and *devoir* both scope over negation.²⁷

4.2.3 Material: context extraction, cleaning, and exclusion procedure

For each condition of Force (possibility/necessity), Age (2–3-year-olds/4–5-year-olds/mothers) and Modal (Study 2a: *devoir*, *pouvoir*; Study 2b: *falloir*, *pouvoir*), we established a list consisting of 20 dialogue contexts (except for the 2–3-year-olds's productions, where for *devoir* we could extract only 17 contexts). The final total number of contexts was thus 237 tests, and 60 controls. The lists are available at <https://osf.io/xqn8b/overview>. Dialogue contexts were randomly sampled from the French corpora described in Section 3 for each modal (*pouvoir*, *devoir*, *falloir*), excluding negated cases, epistemic uses and repetitions, and then manually checked. We did not exclude the contexts where speakers used another (non-target) modal in the preceding dialogue; in these cases, we left it as it is, regardless of its force. In cases where the target modal occurred several times in the dialogue, we put several blanks. Note that when reporting the results, we use the label “mother” for simplicity; however, some utterances tested in the experiment on mother's production come from other adults involved in

²⁷Dieuleveut et al. (2022) also test negated root modal utterances (*can't* vs. *don't have to*) and epistemic modals (*might* vs. *must*). We could not test these in the current study given that epistemic uses are too rare in French children's productions and *avoir à* is the only necessity modal that scopes under negation in French (see footnote 17).

the conversations, such as the investigator and family members (see the list in footnote 10). In the experiment display, we indicated “AUTRE ADULTE” (“other adult”) for all conversational participants who were not the mother or the dad. For the contrast tested in Study 2a (*pouvoir* vs. *devoir*), it was straightforward to interchange *pouvoir/devoir* since their syntactic behavior is identical (see Section 2.3). To interchange *falloir* and *pouvoir* in Study 2b, we proceeded as follows: (i) when *falloir*’s complement is an infinitival (e.g., “Il faut [V+INF],” e.g., “il faut manger”), we use French impersonal subject *on* (participants have to pick between “il faut” and “on peut”); when *falloir* takes a CP complement (e.g., “il faut [_{CP} que tu manges]”), participants have to pick between “tu peux manger” and “il faut que tu manges”; when *falloir*’s expletive subject *il* is dropped,²⁸ we insert it back in the answers (e.g., when the original utterance is “faut manger,” we use “__ manger” so that participants have to pick between “il faut” and “on peut”). For possibility modal contexts, we used the same contexts in Study 2a and 2b, expecting no difference in accuracy. For controls, as in Dieuleveut et al.’s (2022) study, participants had to pick between future and past (e.g., [*a vu*] vs. [*va voir*], “saw” vs. “will see”). Extraction procedure and data cleaning were the same for controls and targets. Importantly, the controls ensured that correct answers were not inferable based on the target utterance alone, requiring participants to read the entire dialogue. We excluded participants who were less than 75% accurate on controls.

4.3 Procedure

Experiments were coded using PennController for Ixex (Zehr and Schwarz 2018) (<https://www.pcibex.net/>) and hosted on the LLF IxexFarm server (<https://ibex.llf-paris.fr/>). Participants first saw instructions (available in the Online Supplemental Appendix B), then three training items with feedback using *le* “the” *un* “a.” Overall, each participant had 40 dialogs to judge: 20 test cases (10 possibility and 10 necessity) and 20 controls using tense (past/future). Test cases were randomly selected (within the PCibex code) out of the list of 20 contexts per condition described in Section 3.2. Dialogs were presented in a randomized order. Participants had to press the space button for each utterance of the dialogue to appear incrementally. The mean duration of the experiment was about 12 minutes. At the end, participants were asked if they had any comments about the task. Demonstrations of the experiments are available at <https://farm.pcibex.net/r/qAcSFz/> (Study 2a, 2- to 3-year-olds); <https://farm.pcibex.net/r/XHSEJM/> (Study 2a, mothers).

4.4 Participants

A total of 358 adult French native speakers were recruited via Prolific (6 groups, 60 participants per group; 2 failed to record data) (166 female, 186 male, 6 non-binary, 1 unknown; M age = 32.8 years, range = 18–74). Exclusion criteria were based on native language (French) and age (over 18-year-old). Participants were paid €2 for their participation. We excluded 11 participants whose accuracy scores on controls were <75% (3.1%). We thus report results from 347 participants (Study 2a: testing mothers’ productions: 59, 2–3-year-olds’: 57; 4–5-year-olds’: 59; Study 2b: mothers’: 59; 2–3-year-olds’: 55; 4–5-year-olds’: 58).

4.5 Analyses

Data analyses were conducted using R (R Core Team 2013). R scripts for analysis are available at <https://osf.io/xqn8b/overview>. To analyze participants’ responses, we fitted a generalized linear mixed-effects model (GLMM) built with a maximal random effect structure allowed by our experimental design (following Barr et al. 2013), implemented with the lme4 package (Bates et al. 2015) to predict

²⁸This occurs 49% of the times in adult speech, 68% in 2- to 3-year-olds speech and 45% in 3- to 6-year-olds speech. See Appendix A for details.

Accuracy (dependant variable, 1 vs. 0, binomial) with Force (possibility vs. necessity), Age (2–3-year-olds vs. 4–5-year-olds vs. mothers), and Modal contrast (Study 2a: *devoir/pouvoir* vs. Study 2b: *falloir/pouvoir*) and their interactions as fixed effects. Random intercepts were included for both Subject and Item. Accuracy was coded as 1 when participants answered with a possibility modal (*pouvoir*) in a possibility context or with a necessity modal (*devoir/falloir*) in a necessity context, and 0 otherwise (glmer syntax of the full model: $\text{Accuracy} = 1 \sim \text{Force} * \text{Modal} * \text{Age} + (1 | \text{Subject}) + (1 | \text{Item})$). We compare the full model to a reduced model without the interaction using a likelihood ratio test (reduced model: $\text{Accuracy} = 1 \sim \text{Force} + \text{Modal} + \text{Age} + \text{Force}:\text{Age} + \text{Force}:\text{Modal} + \text{Age}:\text{Modal} + (1 | \text{Subject}) + (1 | \text{Item})$).²⁹ Categorical predictors were coded using sum contrasts; for the three-level factor Age, contrasts compared each child group to the grand mean, with adults coded as the reference level (–1, –1). To fix convergence issues, we used the “bobyqa” optimizer in R (glmerControl(optimizer=“bobyqa,” calc.derivs = FALSE)). To further investigate the results, we used estimated marginal means (EMMs) with the emmeans package in R (Lenth 2024), looking at simple effects of Force in each Age × Modal group, Age in each Force × Modal and Modal in each Force × Age group, with the Holms correction to control for multiple comparisons.

4.6 Results

Accuracy on controls was high (91.2%) (Study 2a, mothers: 96.1%; 2–3-year-olds: 88.1%; 4–5-year-olds: 90.8%; Study 2b: mothers: 96.0%; 2–3-year-olds: 87.6%; 4–5-year-olds: 91.8%). Figure 4 plots the mean accuracy for test cases by Force and Age in Studies 2a and 2b; Table 4 provides the corresponding percentages accuracy with se. In the experiments testing mothers’ production (baseline), we find (as in English) that participants are highly accurate at guessing modal force, with no difference between possibility and necessity contexts (general mean accuracy: 77.6%; on possibility contexts: 77.9%; on necessity contexts: 77.4%). In the experiments testing child productions, we see, again as in

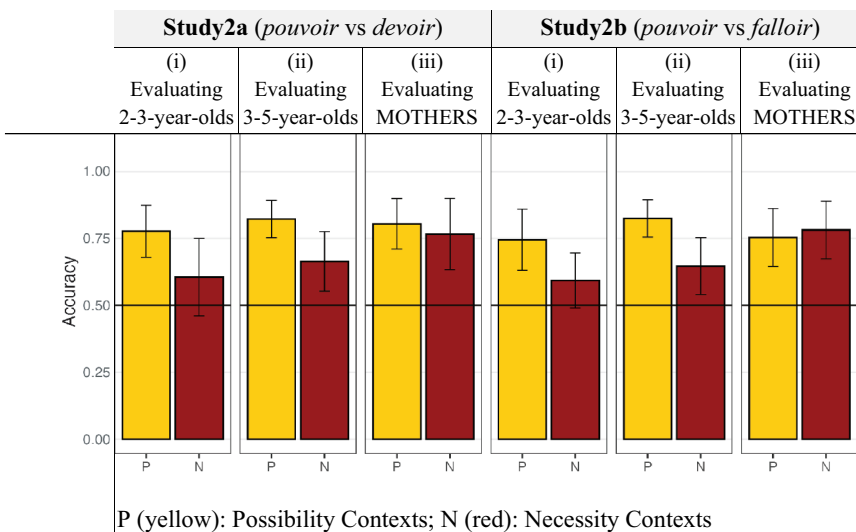


Figure 4. Mean accuracy at recovering the original modal by age and force, Study2a (*pouvoir* vs. *devoir*) and Study2b (*pouvoir* vs. *falloir*). (i) Experiment testing children aged 2 to 3’s production. (ii) Children aged 4 to 5’s production. (iii) Mothers’ production. ($n = 347$ participants). Error bars represent 95% confidence intervals.

²⁹We initially specified a maximal random-effects structure including +Force|Subject but this model resulted in a singular fit (perfect correlation between random intercepts and slopes). We therefore simplified the random-effects structure to include random intercepts only.

English, a difference between possibility and necessity, with higher performance on possibility than necessity contexts, for both Study 2a and 2b, and for both younger and older children (mean accuracy: 2–3-year-olds: possibility: 76.1%; necessity: 59.8%; 4–5-year-olds: possibility: 82.4%; necessity: 65.5%).

Post hoc tests against chance level (0.5) using the `test()` function in `emmeans` are reported in Table 5a. They show that performance was significantly above chance in all adult conditions and children possibility conditions, but not in child necessity conditions. Results of the model testing the effect of the three-way interaction Force*Age*Modal reveal no significant effect (result of the likelihood ratio test comparing models with and without the interaction: $\chi^2(2) = 0.49$, $p = .79$). This indicates that the pattern of effects does not differ between the two modals (*devoir* vs. *falloir*). We conducted pairwise comparisons to investigate simpler effects of Force, Age, and Modal using the `emmeans` package (Lenth 2021). Results from comparison tests are summarized in Tables 5b–5d (Table 5b: effects of Force; Table 5c: effects of Modal; Table 5d: effects of Age group). They reveal a consistent effect of Force for child utterances: participants were significantly less accurate in necessity contexts than in possibility contexts for both younger (2–3 years) and older (4–5 years)

Table 4. Percentage mean accuracy (SE) by age and force, Study 2a and 2b.

	Evaluating 2–3-year-olds		Evaluating 4–5-year-olds		Evaluating MOTHER	
	POSS	NEC	POSS	NEC	POSS	NEC
Study2a	77.7% (0.049)	60.5% (0.074)	82.3% (0.036)	66.4% (0.057)	80.4% (0.048)	76.6% (0.068)
Study2b	74.5% (0.058)	59.3% (0.053)	82.5% (0.036)	64.6% (0.054)	75.3% (0.055)	78.2% (0.055)
ALL	76.1% (0.038)	59.8% (0.044)	82.4% (0.025)	65.5% (0.039)	77.9% (0.036)	77.4% (0.043)

The accuracy corresponds to how good participants are to recover correctly the force of the modal originally uttered, given the context. The mean is computed across the 20 contexts initially extracted from the corpus for each condition of force and age. On average, each context was seen by 38.9 participants, ranging between 19 and 59 times. POSS: possibility; NEC: necessity.

Table 5a. Estimated marginal means (EMMs) by age, modal, and force, Study 2.

Age	Lemma	Force	emmean	SE	Z	p value
MOTHER	Study 2a	POSS	2.144	0.389	5.509	<.0001***
		NEC	1.705	0.385	4.429	.0001***
	Study 2b	POSS	1.625	0.378	4.301	.0001***
		NEC	1.910	0.385	4.965	<.0001***
CHI 2to3	Study 2a	POSS	1.868	0.384	4.864	<.0001***
		NEC	0.662	0.399	1.661	.1900 (NS)
	Study 2b	POSS	1.624	0.381	4.265	.0001***
		NEC	0.514	0.367	1.403	.1900 (NS)
CHI 4to5	Study 2a	POSS	2.079	0.383	5.427	<.0001***
		NEC	0.892	0.369	2.419	.0623 (NS)
	Study 2b	POSS	2.016	0.382	5.282	<.0001***
		NEC	0.867	0.369	2.351	.0623 (NS)

POSS: possibility; NEC: necessity.

Table 5b. Estimated pairwise contrasts for the effect of force (possibility vs. necessity) within each age group and modal type (log-odds scale). Positive values indicate higher accuracy in possibility than necessity. p -values are adjusted using Holm correction.

Age group	Modal	Estimate	SE	z	p
MOTHER	Study 2a	+0.44	0.55	0.80	.422
	Study 2b	−0.29	0.54	−0.53	.596
CHI 2–3	Study 2a	+1.21	0.55	2.18	.029*
	Study 2b	+1.11	0.53	2.10	.036*
CHI 4–5	Study 2a	+1.19	0.53	2.23	.026*
	Study 2b	+1.15	0.53	2.16	.030*

Table 5c. Estimated pairwise contrasts for the effect of modal (Study 2a: *pouvoir* vs. *devoir* vs. Study 2b: *pouvoir* vs. *falloir*) within each age group and force condition (log-odds scale). Positive values indicate higher accuracy in Study 2a relative to Study 2b. *p*-values are adjusted using Holm correction.

Age group	Force	Estimate	SE	z	p
MOTHER	POSS	+0.52	0.54	0.96	.338
	NEC	−0.21	0.54	−0.38	.706
CHI 2–3	POSS	+0.24	0.54	0.45	.652
	NEC	+0.15	0.54	0.27	.785
CHI 4–5	POSS	+0.06	0.54	0.12	.907
	NEC	+0.03	0.52	0.05	.961

POSS: possibility; NEC: necessity.

Table 5d. Estimated pairwise contrasts for the effect of age within each Study and force condition (log-odds scale). *p*-values are adjusted using the Holm correction.

Modal	Force	Contrast	Estimate	SE	z	p
Study 2a	POSS	MOT vs. CHI2–3	+0.28	0.55	0.51	1.000
		MOT vs. CHI4–5	+0.07	0.55	0.12	1.000
		CHI2–3 vs. CHI4–5	−0.21	0.54	−0.39	1.000
	NEC	MOT vs. CHI2–3	+1.04	0.55	1.88	0.180
		MOT vs. CHI4–5	+0.81	0.53	1.52	0.255
		CHI2–3 vs. CHI4–5	−0.23	0.54	−0.42	0.672
Study 2b	POSS	MOT vs. CHI2–3	+0.00	0.54	0.00	1.000
		MOT vs. CHI4–5	−0.39	0.54	−0.73	1.000
		CHI2–3 vs. CHI4–5	−0.39	0.54	−0.73	1.000
	NEC	MOT vs. CHI2–3	+1.40	0.53	2.63	0.026*
		MOT vs. CHI4–5	+1.04	0.53	1.96	0.101
		CHI2–3 vs. CHI4–5	−0.35	0.52	−0.68	0.497

POSS: possibility; NEC: necessity.

Table 5e. Age x Force interaction contrasts (log-odds scale).

Comparison	Estimate	SE	z	p
MOT vs. CHI2–3	1.08	0.54	1.99	.046*
MOT vs. CHI4–5	1.09	0.54	2.03	.042*
CHI2–3 vs.. CHI4–5	0.01	0.54	0.02	.986

children in both studies 2a and 2b. By contrast, no effect of Force was observed when evaluating mothers' utterances, indicating similar performance across possibility and necessity contexts for adult productions (see Table 5b). We find no effect of Modal, regardless of the group tested (see Table 5c). No age differences are found in possibility contexts, whereas for necessity contexts, we find a significant effect when comparing 2–3-year-olds to mothers in Study 2b (see Table 5d). Finally, to assess whether the effect of Force differed across age groups, we tested the Age x Force interaction (likelihood ratio test comparing full model: Accuracy ~ Force * Age + (1|Subject) + (1|Item) vs. reduced model: Accuracy ~ Force + Age + (1|Subject) + (1|Item)). The model comparison reveals a marginal effect ($\chi^2(2) = 5.34$, $p = .069$). Follow-up analyses using estimated marginal means (see Table 5e) show that the effect of Force differ significantly between adults and children. Specifically, the difference between possibility and necessity contexts was significantly larger for 2–3-year-olds than for adults ($z = 1.99$, $p = .046$) and for 4–5-year-olds than for adults ($z = 2.03$, $p = .042$). No difference was found between the two child groups ($z = 0.02$, $p = .99$). Following a reviewer's suggestion, we conducted additional analyses comparing our data (2–3-year-olds and mothers) with those of Dieuleveut et al. (2022) (Experiments 1 and 5, conditions Root-AFF-1 ad Root-AFF-2), which indicate no effect of Language. Detailed results and the R script for these analyses are available in the Online Supplemental Appendix D.

4.7 Discussion

We find high accuracy (i.e., high agreement rates with the child's original choice) when participants had to identify children's possibility modals (on average, 76% for 2–3-year-olds's productions, 82% for 4–5-year-olds,' and 80% for mothers'). This shows that French children, like their English peers, use possibility modals in situations where adults do the same. But when participants had to identify children's necessity modals, we find significantly lower accuracy (i.e., lower agreement rates with the child's choice): 60% for 2–3-year-olds's productions, 66% for 4–5-year-olds,' versus 77% for mothers. This shows that French-speaking children as well “over-generalize” (or “over-use”) necessity modals: they sometimes employ them in contexts where adults would rather use possibility modals. The baseline on mothers' productions, where we find no difference in accuracy between possibility and necessity modals, shows that this is not because necessity modals are intrinsically harder to guess from context. The experiment focusing on the productions of older children (aged 4 to 5) shows that although accuracy increased slightly, children still do not use necessity modals in a fully adult-like manner at this age. Interestingly, we do not find any difference between the two French necessity modals *falloir* and *devoir* (2–3-year-olds: *devoir*: 60.5%; *falloir*: 59.3%; 4–5-year-olds: *devoir*: 66.4% *falloir*: 64.6%), despite their substantial difference in input frequency.

Contexts in (11) and (12) provide examples which led to particularly low accuracy in the experiment, illustrating some children's non-adult-like uses of necessity modals. The example in (13) illustrates a 4-year-old using *falloir*.

- (11) MAMAN : ah! AUTRE ADULTE : ah en dessin alors. MAMAN : et alors tu y arrivais bien? ENFANT : oui! MAMAN : c'est ce que t'as fait aujourd'hui? ENFANT : oui! MAMAN : ah! ENFANT : et puis [**il faut**/on peut] attraper des papillons dans ... dans un filet à papillons.
MUM: ah! OTHER ADULT: Oh, drawing then. MUM: So you were good at it? CHILD: Yes! MUM: Is that what you did today? CHILD: Yes! MUM: Oh! CHILD: And then [you **have to**/you can] catch butterflies in a butterfly net. The bold font indicates the modal originally used by the child.

Paris Corpus, Madeleine, 3;00,28, mean accuracy: 17.4% ($n_{\text{obs}} = 23$)

- (12) MAMAN : alors tu commences par lequel Madeleine? ENFANT : par celui-là. MAMAN : c'est lequel celui-là. ENFANT : euh Elmer. MAMAN : Elmer. MAMAN : c'est ... c'est en hiver? ENFANT : maman là c'est ... euh c'est ça. ENFANT : donc je [peux/**dois**] faire ce puzzle.
MUM: So, which one are you starting with, Madeleine? CHILD: That one. MOM: Which one is that one? CHILD: Uh, Elmer. MUM: Elmer. MUM: Is it ... is it in the winter? CHILD: Mum, it's ... uh, it's this one. CHILD: So I [can/**must**] do this puzzle.

Paris Corpus, Madeleine 2;06,10, mean accuracy: 29.0% ($n_{\text{obs}} = 31$)

- (13) ENFANT : (rit) MAMAN : alors qu'est-ce qu'il y a dans cette ferme tu m'expliques? ENFANT : il y a ... il y a trois ... il y a trois cochons ... ENFANT : et regarde. ENFANT : ... MAMAN : oh là là, tout ça! ENFANT : hum! ENFANT : et regarde comment [**il faut**/on peut] jouer avec.
CHILD: (laughs) MUM: So, what's in this farm can you explain it to me? CHILD: There are ... there are three ... there are three pigs ... CHILD: And look. CHILD: ... MUM: Oh wow, all that! CHILD: Hmm! CHILD: And look how [you **have to**/you can] play with it.

Paris Corpus, Anae, 4;00,12, mean accuracy: 13.8% ($n_{\text{obs}} = 29$)

However, it could be that the lower agreement rate for necessity modals is due do adult participants' expectations would have for children to use possibility modals more often, therefore not necessarily an indication that children are misusing them. When unsure, experimental participants would by default choose a possibility modal if the speaker is a child; This strategy would inflate accuracy for possibility modals while lowering it for necessity modals. To control for this, we ran a follow-up study where we switched the roles of child and mother in the dialogs.

5 Study 3: A follow-up experiment to test for speaker roles

5.1 Goals

To test whether lower performance on children's necessity modals reflects expectations participants might have for children to use more possibility modals—rather than children's actual misuses—we ran a follow-up study where we switched the roles of child and mother, that is, where we attribute what were *children's* utterances to mothers (CHI-to-MOT) and vice versa (MOT-to-CHI). The rationale behind this manipulation is that if participants' choices heavily depend on who they consider to be the speaker, their responses should be affected; whereas, if they mostly rely on the conversational context, their responses should remain identical to those observed in Study 2. More precisely, if participants' lower agreement rates with children's necessity modals in Study 2 (on average 60%) comes from “default” expectations for children to use more possibility modals, we should obtain higher agreement rates if the speaker they see is the mother (reflecting more necessity guesses); For children's possibility modals, we expect lower agreement rates. Conversely, for mothers' utterances (where in Study 2 we find on average 80%, with no difference between possibility and necessity contexts), we should obtain lower agreement rates for necessity modal utterances when attributed to their child (reflecting more possibility guesses), and higher agreement rates for possibility modal utterances.

5.2 Methods and materials

From the participants' perspective, Study 3 was identical to Study 2. We had four groups, varying speakers' age (mother vs. child) and Modal (Study 3a: *pouvoir* vs. *devoir*; Study 3b: *pouvoir* vs. *falloir*). In each group, half of the trials (5/10 possibility, 5/10 necessity, 10/20 controls) tested the dialogs keeping the original speakers (allowing us to replicate Study 2's results); the other half had the child/mother switched (which we abbreviate MOT-to-CHI, CHI-to-MOT). In all groups, participants were judging sentences attributed either to mothers or to children. Figure 5 summarizes the manipulation. Aside from the speaker's labels switch, we did not make any change to the dialogs, except two agreement markings to match gender of the mother in two control contexts (“je vais aller” vs. “je suis allée”).

5.2.1 Context selection

We excluded contexts if which role-switching led to excessively awkward or inappropriate dialogs, as illustrated in (14) and (15). To determine which contexts to exclude, we asked three native French speakers to rate every context from Study 2 on a scale between 1 and 4 (“Does the dialogue sound normal or strange?”; 1: normal; 4: weird). We kept contexts that yielded a mean score below 3. This allowed us to keep 47% of the 177 contexts tested in Study 2 (from MOT: 37/80 (46%); from CHI: 36/77 (47%)). Details are available in the Online Supplemental Appendix D.

- (14) [...] ENFANT : ce que tu peux faire c'est mettre les toilettes dedans. ENFANT : comme ça. ENFANT : hopoo. MAMAN : je ____ faire pipi.
 [...] CHILD: What you can do, is put toilets inside. CHI: like that. CHI: hopoo.
 MUM: I ____ pee.

(Paris corpus, Madeleine, 2;02,06)

- (15) [...] MAMAN : je veux manger dans ma chaise. ENFANT : tu sais aller dans la chaise toute seule? MAMAN : c'est toi tu ____ mettre dans ma chaise.
 [...] MUM: I want to eat in my chair. CHILD: You know how to go in your chair on your own? MUM: It's you you have to put me in my chair.

(Paris corpus, Julie, 2;01,22)

5.2.2 Procedure

The procedure was identical to Study 2. Experiments were coded using PennController for Ibex (<https://www.pcibex.net/>) and hosted on the LLF IbexFarm server (<https://ibex.llf-paris.fr/>). After

instructions and a short training, participants had 40 dialogs to judge, presented in randomized order: 20 trials (10 possibility, 10 necessity, randomly selected from the list) and 20 controls based on tense (past/future), for a mean duration of 12 minutes. Upon completion of the procedure, they were asked for comments.

5.3 Participants

A total of 120 adult French native speakers who had not participated in Study 2 were recruited via Prolific (4 groups, 30 per group) (49 female, 66 male, 2 non-binary, 3 not reported; mean age = 32.5 years, range = 20–68). They were paid €2 for their participation. Exclusion criteria were identical to Study 2. We excluded 2 participants due to their low accuracy on controls (<75%). We thus report results from 118 participants (Study 3a: mother: 29; child: 30; Study 3b: mother: 29; child: 30).³⁰

5.4 Analyses

R scripts for analysis of Study 3 are available at <https://osf.io/xqn8b/overview>. To analyze participants' responses, we fitted a generalized linear mixed-effects model (GLMM) with a maximal random effect structure allowed by the design (following Barr et al. 2013) using the lme4 package (Bates et al. 2015) to predict Accuracy (1 vs. 0, binomial) with Force (possibility vs. necessity), Age (child vs. mother), Switch (*noswitch/switch*) and their interactions as fixed effects. Random intercepts included Subject (by Force and Switch) and Item (by Switch) (glmer syntax of the full model: $\text{Accuracy} \sim 1 \sim \text{Force} * \text{Age} * \text{Switch} * \text{Modal} + (\text{Force} + \text{Switch} \mid \text{Subject}) + (1 \mid \text{Item})$).³¹ As in Study 2, Accuracy was coded as 1 if participants answered with a possibility modal in a possibility context or with a necessity modal in a necessity context, and 0 otherwise. Categorical predictors were coded using sum contrasts. We compare the full model to a reduced model using a likelihood ratio test. To fix convergence issues, we used the “bobyqa” optimizer in R (glmerControl(optimizer=“bobyqa,” calc.derivs = FALSE)). To further investigate the effects, we used estimated marginal means (EMMs) with the emmeans package in R (Lenth 2024), with the Holms correction for multiple comparisons.

5.5 Results

Accuracy on controls was high (overall mean accuracy: 93.2%; MOT: 95.2%; MOT-to-CHI: 94.3%; CHI: 91.7%; CHI-to-MOT: 91.8%), with no difference in accuracy between Switch/No-Switch conditions. Table 6 summarizes mean accuracy in each condition of Force, Age and Switch/No-Switch conditions, for Study 3a and 3b. Its first two rows recall results from Study 2a/b, first on all contexts, second including only the contexts that we could include in Study 3. The last two rows report results from Study 3a/b, first on the contexts where there was no switch (MOT and CHI), then on contexts with the switch (CHI-to-MOT and MOT-to-CHI). Figure 6 plots the results, directly comparing the Switch/No-Switch conditions for each group.

Results of the model testing the effect of the interaction Force*Age*Modal*Switch reveal no significant effect overall (result of the likelihood ratio test: $\chi^2(1) = 0.95, p = .33$). Results from further comparison tests using emmeans focusing on the effect of the Switch are summarized in Table 7. They reveal no effect of the switch overall.

³⁰We tested only half as many participants as for Study 2 because we tested only half of the contexts (see context exclusion procedure). Note that this follow-up study was run before extending the paradigm to test 4 to 5-year-olds' productions, since we wanted to exclude this possible explanation before testing productions from the older age group. Therefore, in this section, we use CHI to refer to 2- to 3-year-olds, and MOT for mothers.

³¹We initially specified a maximal random-effects structure including + Switch | Item but this model resulted in a perfect correlation between random intercepts and slopes, so we simplified the model.

Table 6. Mean accuracy (se) by age and force, Study 3a and 3b ($n = 118$), compared to Study 2a/2b.

		Study 2/3a (<i>pouvoir</i> vs. <i>devoir</i>)				Study 2/3b (<i>pouvoir</i> vs. <i>falloir</i>)			
		CHILD		MOTHER		CHILD		MOTHER	
		POSS	NEC	POSS	NEC	POSS	NEC	POSS	NEC
i	Study 2 (all contexts)	77.7% (0.049)	60.5% (0.074)	80.4% (0.048)	76.6% (0.068)	74.5% (0.058)	59.3% (0.053)	75.3% (0.055)	78.2% (0.055)
ii	Study 2 (kept for Study 3)	80.4% (0.063)	66.1% (0.085)	79.6% (0.070)	72.5% (0.112)	78.1% (0.087)	56.8% (0.088)	73.2% (0.087)	79.0% (0.055)
iii	Study 3 (no switch)	81.8% (0.061)	64.8% (0.105)	79.4% (0.054)	71.4% (0.099)	75.7% (0.088)	59.0% (0.095)	70.0% (0.095)	84.8% (0.039)
iv	Study 3 (switch)	83.7% (0.048)	65.2% (0.094)	73.7% (0.074)	64.4% (0.109)	79.2% (0.068)	52.8% (0.102)	62.4% (0.085)	83.0% (0.028)

POSS: possibility; NEC: necessity.

On average, each context was seen 41 times in Study 3 (20.9 times with original speaker, 20.2 times with switched speakers), ranging between 16 and 60 times.

Table 7. Summary of pairwise contrasts for accuracy (log-odds scale) testing the effect of Switch (no-switch vs. switch), by force condition and (original) age group in Study 3.

Force	Original Age	Study	Estimate	SE	z	p
POSS	MOTHER	Study 3a	+0.366	0.320	1.15	.252
		Study 3b	+0.533	0.310	1.72	.086
	CHILD	Study 3a	−0.175	0.383	−0.46	.648
		Study 3b	−0.294	0.397	−0.74	.460
NEC	MOTHER	Study 3a	+0.501	0.347	1.44	.149
		Study 3b	+0.087	0.364	0.24	.811
	CHILD	Study 3a	−0.008	0.313	−0.03	.979
		Study 3b	+0.314	0.287	1.09	.274

POSS: possibility; NEC: necessity.

Positive values indicate higher accuracy in no-switch trials. p -values are adjusted using Holm correction.

5.6 Discussion

Results from the follow-up shows no effect of the switch in roles. In fact, the only condition where there might be a trend are children's possibility contexts, where accuracy is slightly higher when they are attributed to their mothers. We find the reverse tendency for mother's possibility modal utterances: the switch leads to lower accuracy, suggesting that participants might in fact use more necessity modals when the modal is attributed to the child (in this condition, the effect is marginally significant for Study 3b). If anything, the effect therefore seems to go in the opposite direction of what we would find if participants were merely picking a possibility modal for a child utterance by default. The findings from this follow-up experiment thus strengthen Study 2's conclusions, as it shows that low accuracy on children's necessity modals is not due to participants' expectations about speakers.

6 General discussion

What have we learned by looking at French? Our study has three main takeaways. First, we show that children's difficulties with necessity modals are not specific to English learners: the *Necessity Gap* clearly extends to French. The findings for French and English are actually remarkably similar. In both languages, children master possibility modals (*pouvoir*, *can*, etc.) at a very young age. By age two, they use them frequently, in varied syntactic environments (with and without negation, in declarative and interrogative sentences), and our experiment shows that they use them in an adult-like way, i.e., in the same conversational contexts as adults would. This contrasts with their later mastery of necessity modals: in both languages, children start producing them later, with an average delay of four months; they use them less frequently and in less varied syntactic contexts; and our experiment shows that they sometimes employ them in situations where adult speakers prefer to use possibility modals. Study 3

	No switch	Switch
CHI productions	CHI <i>Child original utterance (Study 2)</i> ENFANT : ... t'en laisses un petit coup. MAMAN : merci. ENFANT : voilà. MAMAN : merci. ENFANT : arrête d'aller là avec le ptit chevaux ENFANT : vous arrêtez d'aller là. ENFANT : parce que c'est après. ENFANT : qu'on _____ aller après. doit peut	CHI-to-MOT <i>Child utterance attributed to Mother</i> MAMAN : ... t'en laisses un petit coup. ENFANT : merci. MAMAN : voilà. ENFANT : merci. MAMAN : arrête d'aller là avec le ptit chevaux MAMAN : vous arrêtez d'aller là. MAMAN : parce que c'est après. MAMAN : qu'on _____ aller après. doit peut
	CHILD: ... you leave a little bit. MUM: thank you. CHILD: there you go. MUM: thank you. CHILD: stop going there with the little horses. CHILD: stop going there. CHILD: because it's later. CHILD: that we _____ go later.	MUM: ... you leave a little bit. CHILD: thank you. MUM: there you go. CHILD: thank you. MUM: stop going there with the little horses. MUM: stop going there. MUM: because it's later. MUM: that we _____ go later.
MOT productions	MOT <i>Mother's original utterance (Study 2)</i> AUTRE ADULTE : oui. MAMAN : oui AUTRE ADULTE : une soucoupe. ENFANT : sont un peu vieilles. MAMAN : oui sont un peu abîmées tordues. MAMAN : ah celle-là elle marche bien. AUTRE ADULTE : merci beaucoup. MAMAN : tu _____ souffler dessus. doit peut	MOT-to-CHI <i>Mother utterance attributed to Child</i> AUTRE ADULTE : oui. ENFANT : oui AUTRE ADULTE : une soucoupe. MAMAN : sont un peu vieilles. ENFANT : oui sont un peu abîmées tordues. ENFANT : ah celle-là elle marche bien. AUTRE ADULTE : merci beaucoup. ENFANT : tu _____ souffler dessus. doit peut
	OTHER ADULT: yes. MUM: yes. OTHER ADULT: a saucer. CHILD: they're a bit old. MUM: yes, they're a bit damaged, bent. MUM: ah, that one works well. OTHER ADULT: thank you very much. MUM: you _____ blow on it.	OTHER ADULT: yes. CHILD: yes. OTHER ADULT: a saucer. MUM: they're a bit old. CHILD: yes, they're a bit damaged, bent. CHILD: ah, that one works well. OTHER ADULT: thank you very much. CHILD: you _____ blow on it.

Figure 5. Summary of the manipulation: example trials without/with the switch. Each participant saw either contexts from CHI and MOT-to-CHI conditions (light grey), or from MOT and CHI-to-MOT conditions (dark grey). Abbreviations: CHI, child; MOT, mother.

shows that this is not just an experimental artifact coming from adult participants expecting young children to use by default more possibility modals. Second, our study shows that these difficulties persist with older children—a point that had not been assessed by Dieuleveut et al. (2022). We find slightly higher performance on 4- to 5-year-olds' production, but they still do not use their necessity modals in a fully adult-like way.

While within the broader literature on early lexical acquisition, there is substantial variation in the methods used to assess children's language skills, and debates regarding what the most valid measure is

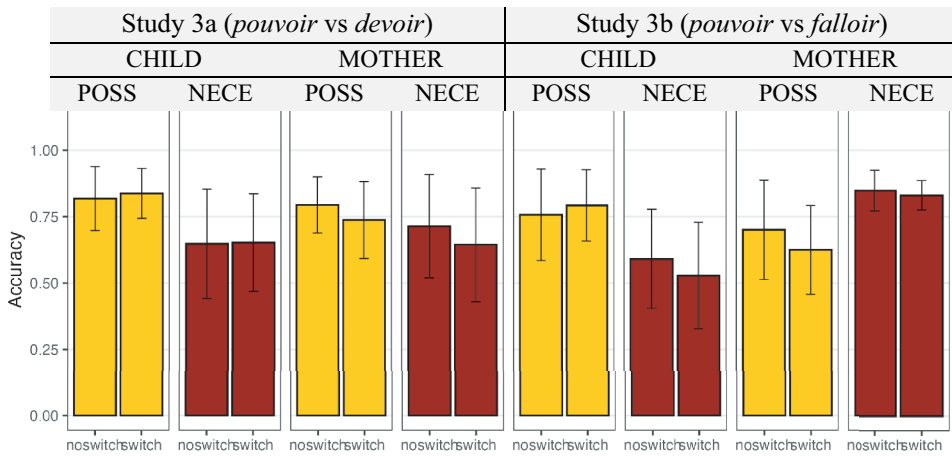


Figure 6. Effect of role switch, Study 3a and 3b. Error bars represent 95% confidence intervals.

(see, e.g., Bornstein and Haynes 1998; Hoff 2006; Rowe 2012; Ambridge et al. 2015), one strength of our study is that we combine three indicators of children’s mastery of modals: frequency of usage from age 2 to 6, age of first production, and whether they spontaneously use these words in similar conversational contexts as adults. In particular, the experimental paradigm we borrow from Dieuleveut et al. (2022) allows to go beyond pure quantitative data and assess children’s natural spontaneous production (children were recorded at home, in their natural environment, in interaction with their parents, which might be seen as more ecologically valid than lab assessment). Additionally, it is simple and easy to implement, requiring minimal demands from participants; since it is based on existing corpus data, it can be readily generalized to other languages and allows direct crosslinguistic comparisons.

Of course, looking at production is limited; it is thus important to highlight that our results align with results from comprehension studies. As we reviewed in Section 2.2., these studies show that 4–5-year-olds, and even older children, both overaccept possibility modals in necessity situations, and necessity modals in possibility situations, when tested in epistemic as well as root modality (Noveck et al. 1996; Noveck 2001, Öztürk and Papafragou 2015; Moscati et al. 2017; Leahy and Zalnieriunas 2021; Staniszewski et al. 2023; Bleotu et al. 2025; Cournane et al. 2025). Recent work suggests that these comprehension results extend to French as well: in a similar comprehension task as Cournane et al.’s (2025), French-speaking 4- to 5-year-olds behave in an adult-like way for *pouvoir*, but tend to accept “Il faut/Tom doit prendre la route jaune” “Tom must take the yellow road” in situations where two roads are open (see Lungu et al. 2026).

Our third, important finding, allowed by the comparison between French and English, is that limited exposure cannot explain the *Necessity Gap*. When comparing French and English children’s input, we find that French parents use necessity modals much more than possibility modals, and more than English parents do. Despite this greater exposure, French children experience the same difficulties with necessity modals. Additionally, focusing on French, we find no evidence for *falloir* being mastered earlier than *devoir*, despite *falloir* being over six times much more frequent than *devoir* in the input. These are remarkable results, since input quantity is a strong predictor of lexical acquisition (Anderson et al. 2021): all else equal, more frequent words are expected to be acquired earlier. Our study shows that in this case, other mechanisms must be at play.

Our study demonstrates that the delay for necessity modals cannot be attributed to limited exposure. If this is not about quantity of exposure, why do children struggle with necessity modals? As of now, we cannot empirically distinguish the different hypotheses for children struggles with

necessity modals, reviewed in [Section 2.2](#). However, our study opens promising directions, as it rules out one potential explanation for children's difficulties—insufficient exposure.

The first key question to address next is whether these results do or do not extend to other cases of word learning: nouns such as *fruit/plums*, as well as other logical Horn scales, such as *some/all* or *sometimes/always*, where similar *subset problems* arise. Do children have issues with logically stronger terms (*apple, all, or always*), and do they “overgeneralize” their uses? If they do, that might affect the kind of explanation we want to give to *Necessity Gap*. Interestingly, children's overgeneralization errors (or “overextensions”) have been reported and quite well studied in the domain of content words (e.g., Clark 1973; Bowerman 1978; Rescorla 1980; Ambridge et al. 2013; Hoff 2006). The literature on the topic generally distinguishes between two types of errors: overgeneralization errors—when a child uses a word to refer to too wide a category (e.g., *dog* to refer to a bear, *orange* to refer to some other fruit), and undergeneralization errors (“underextensions”)—when a child for instance uses a word to refer to too small a category (e.g., *dog* for a specific pet only).³² Overextensions are well attested in natural productions, as well as in more controlled naming tasks (e.g., Duvignau et al. 2007) and categorization tasks (e.g., Schaefer 1979); however, there is a debate regarding what underlies such errors. They have been taken to be conceptual category errors, pragmatic adaptations to vocabulary limitations, or taken to reflect young children's analogical abilities (see e.g., Raynal et al. 2024); however, it may also be that children assume that weaker terms have a more general meaning (e.g., that *orange* means FRUIT).

One challenge for the hypothesis that children have conceptual difficulties *representing* possibilities as argued by Leahy and Carey (2020) (see, also, Leahy and Zalnieriunas 2021; Leahy 2023) is to explain children's remarkably young mastery of possibility modals shown by their spontaneous natural modal productions.³³ To explain their results on English, Dieuleveut et al. (2022) propose that children may have not (yet) figured out the underlying force of their necessity modals, potentially due to the complexity of the word-to-concept mapping that modals pose (the *subset problem*) (see also Cournane et al. 2025). In the strong version of their hypothesis, young children think that modals such as *falloir/have-to* express possibility. According to Cournane et al. (2025), this could reflect a general possibility meaning bias for modals. While this account may capture the findings from both production and comprehension, it requires answering an additional question: how do children eventually figure out the meaning of necessity modals? If children's difficulties turn out to be cross-linguistically consistent—which remains to be established, although our study takes a first step in this direction—this will call for a general solution. Dieuleveut (2021) argues that children cannot rely on negative environments (Gualmini and Schwarz 2009),³⁴ but proposes that they might use contextual solve the problem, notably cues stemming from desirability. Is this enough, and should we expect such contextual cues to be available in every language? Finally, while less developed, another line of explanation underlines pragmatic immaturity (Dieuleveut 2021). Children would know the meaning of necessity modals, have the right underlying concepts, but fail to use (and understand) necessity modals in appropriate contexts because of difficulties

³²Note that undergeneralization errors may be much harder to detect. When a child overextends, parents should easily be able to notice, which might also be the reason why these errors have so much been studied, in the domain of content words. By contrast, if a child underextends, parents do not have direct evidence that they do. The *subset problem* is here reversed. Also note that for content words' overgeneralizations, parents may be able to provide direct “corrective” (explicit) feedback (e.g., “it's not a dog, it's bear,” which may tell the child that the meaning of *dog* is not ANIMAL). But while this kind of feedback might help solving *subset problems* for content words, it may not be so straightforward to exploit with modals, since to be able to provide such direct feedback, parents have to realize that the child did not intended to express necessity while using a necessity modal.

³³As pointed out by an anonymous reviewer, our experiment could however give false positives: children could use the words appropriately without knowing their meaning. An analogy from number acquisition illustrates this concern. It is known that children first learn a counting procedure, before actually knowing the meaning of number words like *five* or *six*. A “1-knower,” asked to count an array of 5 items, will correctly recite the counting sequence. If we blank out the last number word the child says, and ask adults to fill in the blank from two options (e.g., five and six), accuracy will presumably be very high. However, such performance would overestimate the child's competence, as 1-knowers do not in fact understand the meaning of *five*.

³⁴As shown by Dieuleveut (2021), if children use negative environments to learn modal's force (a solution proposed by Gualmini and Schwarz (2009) as a principled way to solve semantic *subset problems* in general), a problem arises from the irregularities of modal and negation scope interactions: children might be misled into thinking that necessity modals like *falloir* or *devoir*, *must*, and *should*, express possibility (see footnote 17). Note that interestingly, it has been shown that children can use negative evidence to learn the meaning of novel words in the case of content words (see de Carvalho et al. 2021; de Carvalho and Dautriche 2025).

determining the intended modal domain—that is, the range of available options in a given situation. This hypothesis deserves to be explored and tested further.

It remains crucial to extend the empirical findings to other languages, where modal systems differ. This will permit to see whether as expected, the *Necessity Gap* replicates beyond English and French. In this respect, we believe that modal systems where scope interactions between modals and negation are better behaved (if there exists any), would be of particular interest. Within French, examining how children acquire modal expressions such as *être obligé/forcé à/de* (“be required/forced to”) will also be informative, which contrasts in force with *être autorisé à* (“be allowed to”). While initial searches show that these expressions are less frequent, several factors may make their acquisition easier: they do not show flavor variability, and they are interpreted below negation.

7 Conclusion

This study makes some important steps in our understanding of children’s difficulties with necessity modals. We have shown that children’s difficulties with necessity modals are not specific to English learners and extend to French. Crucially, the comparison between French and English shows that limited quantitative exposure cannot explain the asymmetry found for English: despite a higher exposure to necessity modals, French children too encounter difficulties with necessity modals, with no difference between *falloir* and *devoir*. What remains to be determined is whether these difficulties reflect semantic difficulties related to the complexity of the word-to-concept mapping problem that modals pose, conceptual difficulties with possibilities, or pragmatic issues. Future research will address these questions.

Acknowledgments

We are very grateful to two anonymous reviewers for their thoughtful and constructive comments, which helped us improve the paper. We further thank Morgan Moyer, Valentine Hacquard, Ailís Cournane, Oana Lungu, Annemarie van Dooren, Keny Chatain, Isabelle Charnavel, Tom Meadows, David Mueller, Milica Denic, and Dominique Juffin for helpful discussions and feedback at various stages of this work. We also thank the members of the GRISP group at the Laboratoire de Linguistique Formelle and of the NYU/UMD ModSquad for their support and input, as well as audiences at BUCLD 47 and ELM 3 for valuable comments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by a Fyssen Foundation post-doctoral study grant and by the Chaire d’Excellence Université de Paris Cité.

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Data availability statement

The datasets, experimental materials and analyses supporting the findings of this study are openly available at: <https://osf.io/xqn8b/overview>. Supplementary materials include:

- (1) Corpus data and transcription guide
- (2) Data and R script analyses for Study 2 & Study 3
- (3) Appendix A: Corpus results, Interaction with subject, Study 1

- (4) Appendix B: Instructions and procedure, Study 2 & 3
- (5) Appendix C: Model R outputs from statistical analyses, Study 2 & 3
- (6) Appendix D: Analyses comparing French to English (Study 2)
- (7) Appendix E: Details about exclusion procedure, Study 3.

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